Geometry II

Key definitions

Definition 1 (Topological surface / 2–manifold). *A* topological surface *is a Hausdorff, second–countable topological space S such that every point $p \in S$ has an open neighbourhood U for which there exists a homeomorphism*

$$\phi : U \longrightarrow V$$

onto some open set $V \subset \mathbb{R}^2$. Equivalently, S is a 2–dimensional topological manifold.

Definition 2 (Cone in $\mathbb{R}^3$). *The (double) cone in $\mathbb{R}^3$ is the subset*

$$C = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 = z^2\}.$$

The point $(0, 0, 0)$ is called the apex *of the cone.*

Definition 3 (Quotient space). *Let Y be a topological space and let $\sim$ be an equivalence relation on Y. The* quotient space *$X = Y/\sim$ is the set of equivalence classes $[y]$ equipped with the* quotient topology*: a set $U \subset X$ is declared open if and only if its preimage*

$$\pi^{-1}(U) \subset Y$$

under the canonical projection $\pi : Y \to X$, $\pi(y) = [y]$, is open in Y.

Definition 4 (Hausdorff space). *A topological space X is* Hausdorff *(or T_2) if for any two distinct points $x \neq y$ in X there exist disjoint open sets $U, V \subset X$ such that $x \in U$ and $y \in V$.*

Definition 5 (Second–countable space). *A topological space X is* second countable *if there exists a countable collection $\mathcal{B}$ of open sets in X such that every open set in X can be written as a union of elements of $\mathcal{B}$. The family $\mathcal{B}$ is then called a* countable base *(or* countable basis*) for the topology.*

Definition 6 (Discrete topology). *Let S be a set. The* discrete topology *on S is the topology in which every subset $U \subset S$ is declared open. In particular, each singleton $\{s\}$ is an open set. The symbol $\mathbb{R}_\delta$ denotes the set $\mathbb{R}$ equipped with the discrete topology.*

Definition 7 (Product topology). *If (X, τ_X) and (Y, τ_Y) are topological spaces, the* product topology *on $X \times Y$ is the topology generated by the basis consisting of all sets of the form*

$$U \times V, \qquad U \in \tau_X,\ V \in \tau_Y.$$

The resulting topological space is denoted $X \times Y$.

Second countability and manifolds

By definition, a topological n–manifold is a Hausdorff space M which is locally Euclidean of dimension n and *second countable*; that is, the topology of M admits a countable base.

The local Euclidean condition ensures that every point has a neighbourhood homeomorphic to $\mathbb{R}^n$. The second countability condition is added to exclude nasty examples and to guarantee good global properties.

Without second countability one can construct spaces which are locally Euclidean but behave very badly globally, for example:

- an uncountable disjoint union of spheres, which is locally a 2–manifold but completely unlike the surfaces usually studied;
- the long line and its higher-dimensional analogues, which are locally like $\mathbb{R}$ or $\mathbb{R}^2$ but are not metrizable, have no countable atlas, and fail many standard theorems.

For manifolds, second countability implies in particular that M is separable, paracompact and metrizable, and that it admits a countable atlas of charts. These properties are used throughout the theory, e.g. in the existence of partitions of unity, Riemannian metrics, and the classification of surfaces. Thus the second countability axiom rules out pathological "huge" manifolds and matches the intuitive notion of a reasonable surface or space.

Problem 4: Cylindrical projection, lunes and spherical triangles

(a) Statement. Place the unit sphere $S^2 \subset \mathbb{R}^3$ inside a vertical circular cylinder C of radius one. Prove that horizontal projection from S^2 to C preserves area. Deduce that S^2 admits a smooth atlas of charts which are area-preserving.

Parametrisations and area element. Use coordinates (θ, z) with $\theta \in (-\pi, \pi)$ and $z \in (-1, 1)$. The unit sphere is parametrised by

$$X(\theta, z) = (\sqrt{1 - z^2} \cos\theta,\ \sqrt{1 - z^2} \sin\theta,\ z),$$

while the cylinder C is parametrised by

$$Y(\theta, z) = (\cos\theta, \sin\theta, z).$$

The horizontal projection $\Pi : S^2 \to C$ satisfies

$$\Pi \circ X(\theta, z) = Y(\theta, z),$$

i.e. it preserves the parameters (θ, z).

For a parametrisation $X(u, v)$ the area element is

$$dA = |\partial_u X \times \partial_v X|\, du\, dv = \sqrt{EG - F^2}\, du\, dv,$$

where $E = X_u \cdot X_u$, $F = X_u \cdot X_v$, $G = X_v \cdot X_v$.

Sphere. Let $r(z) = \sqrt{1 - z^2}$. Then

$$X_\theta = (-r \sin\theta,\ r\cos\theta,\ 0), \qquad r'(z) = -\frac{z}{r}.$$
$$X_z = (r'(z)\cos\theta,\ r'(z)\sin\theta,\ 1),$$

Thus

$$X_z = \left(-\tfrac{z}{r}\cos\theta,\ -\tfrac{z}{r}\sin\theta,\ 1\right).$$

We obtain

$$\begin{aligned} E &= X_\theta \cdot X_\theta = r^2 = 1 - z^2, \\ F &= X_\theta \cdot X_z = 0, \\ G &= X_z \cdot X_z = \frac{z^2}{1-z^2} + 1 = \frac{1}{1-z^2}. \end{aligned}$$

Hence

$$EG - F^2 = (1 - z^2) \cdot \frac{1}{1-z^2} - 0 = 1,$$

and the area element on S^2 is

$$dA_{S^2} = d\theta\, dz.$$

Cylinder. For the cylinder,

$$Y_\theta = (-\sin\theta, \cos\theta, 0), \qquad Y_z = (0, 0, 1),$$

so

$$E' = Y_\theta \cdot Y_\theta = 1, \quad F' = 0, \quad G' = Y_z \cdot Y_z = 1.$$

Therefore

$$dA_C = \sqrt{E'G' - F'^2}\, d\theta\, dz = d\theta\, dz.$$

Area preservation and atlas. Since Π preserves the parameters (θ, z) and both surfaces have area element $d\theta\, dz$ in these coordinates, Π is area-preserving:

$$\mathrm{Area}(\Pi(U)) = \mathrm{Area}(U)$$

for any suitable region $U \subset S^2$ (away from the poles).

On the cylinder the map

$$(\theta, z) \mapsto (\theta, z) \in \mathbb{R}^2$$

is a local isometry onto an open subset of the plane, so it is also area-preserving. For each patch we obtain a chart

$$S^2 \xrightarrow{\Pi} C \xrightarrow{(\theta,z)} \mathbb{R}^2$$

which is area-preserving. Taking finitely many such patches gives a smooth atlas of area-preserving charts on S^2.

(b) Lunes and spherical triangles.

Definition. A *lune* on S^2 is a region bounded by two great circles; its interior angle is the angle α between these circles. A *spherical triangle* is a connected region bounded by three great circles, with interior angles $\alpha, \beta, \gamma < \pi$.

Area of a lune. By applying a rotation of S^2, a lune of angle α can be assumed to lie between the meridians $\theta = \theta_0$ and $\theta = \theta_0 + \alpha$. In (θ, z)-coordinates the lune is

$$\theta_0 < \theta < \theta_0 + \alpha, \qquad -1 < z < 1.$$

Using $dA = d\theta\, dz$,

$$\mathrm{Area}(\text{lune of angle } \alpha) = \int_{\theta_0}^{\theta_0+\alpha} \int_{-1}^{1} 1\, dz\, d\theta = \alpha \cdot 2 = 2\alpha.$$

Area of a spherical triangle (Girard's theorem). Let T be a spherical triangle with angles $\alpha, \beta, \gamma < \pi$. For each angle, form the lune whose boundary circles contain the two sides adjacent to that angle; denote these by $L_\alpha, L_\beta, L_\gamma$. From above,

$$\text{Area}(L_\alpha) = 2\alpha, \quad \text{Area}(L_\beta) = 2\beta, \quad \text{Area}(L_\gamma) = 2\gamma.$$

The three great circles defining T partition the sphere into 8 spherical triangles, among them T and its antipodal copy T'. Each lune is the union of four of these triangles; a counting argument shows that every point of S^2 lies in the union $L_\alpha \cup L_\beta \cup L_\gamma$ exactly once, except that the triangles T and T' are each counted three times. Consequently

$$\text{Area}(S^2) = \text{Area}(L_\alpha) + \text{Area}(L_\beta) + \text{Area}(L_\gamma) - 4\ \text{Area}(T).$$

For the unit sphere, $\text{Area}(S^2) = 4\pi$, so

$$4\pi = 2\alpha + 2\beta + 2\gamma - 4\ \text{Area}(T),$$

and therefore

$$\text{Area}(T) = \alpha + \beta + \gamma - \pi.$$

Examples.

- A lune of angle $\pi/2$ has area $2 \cdot \frac{\pi}{2} = \pi$. Four such lunes cover the unit sphere, giving total area 4π.
- A spherical triangle bounded by the equator and two meridians $90°$ apart has angles $\alpha = \beta = \gamma = \frac{\pi}{2}$, so
$$\text{Area} = \frac{\pi}{2} + \frac{\pi}{2} + \frac{\pi}{2} - \pi = \frac{\pi}{2}.$$
Indeed, four such triangles tile the sphere, so each has area $4\pi/4 = \pi/2$.

Let S^2_R be the sphere of radius R.

- If $p, q \in S^2_R$ are distinct and not antipodal, then there is a unique great circle C containing them. The shortest path in S^2_R from p to q is the *shorter arc* of C between p and q. This arc is the geodesic segment joining p and q.
- If p and q are antipodal (i.e. $q = -p$ on the unit sphere), then every great circle through p and q has an arc of length πR between them. All these arcs are geodesics, and they all have the same length, so the shortest path is not unique.

Remark. Consider the unit sphere S^2 and the following three great circles:

- the equator $\phi = 0$,
- the meridian $\lambda = 0$,
- the meridian $\lambda = \frac{\pi}{2}$.

These bound a spherical triangle T.

At each vertex of T the angle is a right angle:

- where the equator meets $\lambda = 0$, the angle is $\frac{\pi}{2}$,

- where the equator meets $\lambda = \frac{\pi}{2}$, the angle is $\frac{\pi}{2}$,
- at the north pole, the two meridians meet at angle $\frac{\pi}{2}$.

Hence

$$\alpha = \beta = \gamma = \frac{\pi}{2}, \qquad \alpha + \beta + \gamma = \frac{3\pi}{2} = 270^\circ.$$

Each pair of these great circles bounds a lune of angle $\frac{\pi}{2}$, and by the lune formula

$$\text{Area(lune of angle } \alpha) = 2\alpha,$$

we get area π for such a lune. Four congruent triangles T tile the sphere, so

$$\text{Area}(T) = \frac{4\pi}{4} = \frac{\pi}{2}.$$

Girard's theorem for a spherical triangle on the unit sphere states

$$\text{Area}(T) = \alpha + \beta + \gamma - \pi.$$

Substituting $\alpha = \beta = \gamma = \frac{\pi}{2}$,

$$\text{Area}(T) = \frac{\pi}{2} + \frac{\pi}{2} + \frac{\pi}{2} - \pi = \frac{3\pi}{2} - \pi = \frac{\pi}{2},$$

in agreement with the tiling argument. Thus it is perfectly possible for a spherical triangle to have total angle 270°.

For a spherical triangle T on the unit sphere with angles α, β, γ (each $< \pi$), Girard's theorem states

$$\text{Area}(T) = \alpha + \beta + \gamma - \pi.$$

Since $\text{Area}(T) > 0$, we always have

$$\alpha + \beta + \gamma > \pi,$$

so the angle sum of a spherical triangle is strictly larger than π (180°). On the other hand, for a nondegenerate triangle one has

$$\alpha + \beta + \gamma < 3\pi,$$

so the angle sum lies strictly between π and 3π.

In particular, a triangle with three right angles $\alpha = \beta = \gamma = \frac{\pi}{2}$ has angle sum

$$\alpha + \beta + \gamma = \frac{3\pi}{2} = 270^\circ,$$

but this is only one special case. Spherical triangles can have many different angle sums, always strictly between 180° and 540°, depending on their size on the sphere.

Remark. On the sphere the shortest path between two non–antipodal points is the shorter arc of the unique great circle through them; this arc is a geodesic.

A *loxodrome* (or rhumb line) is a curve on the sphere that cuts all meridians under a constant angle. In general, loxodromes are *not* geodesics and therefore do not give shortest paths. They coincide with great circles only in the special cases:

- constant bearing 0° or 180°: meridians;
- constant bearing 90° or 270°: the equator.

Apart from these special cases, a loxodrome between two points on the sphere is strictly longer than the corresponding great–circle segment.

Loxodromes and the Mercator projection

We use geographic coordinates (ϕ, λ) on the unit sphere, where ϕ is latitude and λ is longitude.

Definition of a loxodrome. A *loxodrome* (rhumb line) is a curve on the sphere that cuts all meridians at a constant angle α (measured clockwise from north).

Parametrise the curve by ϕ:

$$\phi \mapsto (\phi, \lambda(\phi)).$$

The spherical metric is

$$ds^2 = d\phi^2 + \cos^2\phi\, d\lambda^2.$$

Along the curve, a tangent vector is

$$T = \dot{\phi}\,\partial_\phi + \dot{\lambda}\,\partial_\lambda,$$

with $\dot{\phi} = d\phi/dt$, $\dot{\lambda} = d\lambda/dt$. The meridian direction is ∂_ϕ.

The angle α between T and ∂_ϕ satisfies

$$\cos\alpha = \frac{\langle T, \partial_\phi \rangle}{\|T\|\,\|\partial_\phi\|} = \frac{\dot{\phi}}{\sqrt{\dot{\phi}^2 + \cos^2\phi\,\dot{\lambda}^2}}.$$

Equivalently, one may use

$$\tan\alpha = \frac{\text{(horizontal component)}}{\text{(vertical component)}} = \frac{\cos\phi\,|\dot{\lambda}|}{|\dot{\phi}|}.$$

Assuming $\dot{\phi} \neq 0$, we obtain the differential equation

$$\frac{d\lambda}{d\phi} = \frac{\tan\alpha}{\cos\phi}.$$

Solving the loxodrome equation. We integrate:

$$\lambda(\phi) = \tan\alpha \int \frac{d\phi}{\cos\phi} = \tan\alpha\, \ln\tan\!\left(\frac{\pi}{4} + \frac{\phi}{2}\right) + C,$$

where we used the standard antiderivative

$$\int \frac{d\phi}{\cos\phi} = \ln\tan\!\left(\frac{\pi}{4} + \frac{\phi}{2}\right) + \text{const.}$$

Mercator coordinates. The Mercator projection is given by

$$x = \lambda, \qquad y = \ln\tan\!\left(\frac{\pi}{4} + \frac{\phi}{2}\right).$$

In terms of (x, y), the loxodrome relation becomes

$$x = \lambda = \tan\alpha\, y + C.$$

This is the equation of a straight line in the (x, y)-plane.

Conclusion. Under the Mercator projection, every loxodrome (curve of constant bearing α on the sphere) is mapped to a straight line

$$x = \tan\alpha\, y + C$$

in the map plane. Conversely, the preimage on the sphere of any non-vertical straight line in Mercator coordinates is a loxodrome.

On the unit sphere with metric

$$ds^2 = d\phi^2 + \cos^2\phi\, d\lambda^2,$$

a loxodrome is a curve $\phi \mapsto (\phi, \lambda(\phi))$ that meets all meridians at a constant angle α. One finds

$$\tan\alpha = \frac{\cos\phi\,|\frac{d\lambda}{d\phi}|}{1},$$

so, up to sign,

$$\boxed{\frac{d\lambda}{d\phi} = \frac{\tan\alpha}{\cos\phi}}.$$

Integrating,

$$\lambda(\phi) = \tan\alpha \int \frac{d\phi}{\cos\phi} = \tan\alpha\, \ln\tan\Big(\frac{\pi}{4} + \frac{\phi}{2}\Big) + C.$$

Geometry of the unit sphere in (ϕ, λ)

Let latitude $\phi \in (-\pi/2, \pi/2)$ and longitude $\lambda \in (-\pi, \pi)$. A point of the unit sphere is

$$S(\phi, \lambda) = (\cos\phi\cos\lambda,\ \cos\phi\sin\lambda,\ \sin\phi).$$

The partial derivatives are

$$\begin{aligned} S_\phi &= (-\sin\phi\cos\lambda,\ -\sin\phi\sin\lambda,\ \cos\phi),\\ S_\lambda &= (-\cos\phi\sin\lambda,\ \cos\phi\cos\lambda,\ 0). \end{aligned}$$

Hence the first fundamental form coefficients are

$$\begin{aligned} E &= S_\phi \cdot S_\phi = 1,\\ F &= S_\phi \cdot S_\lambda = 0,\\ G &= S_\lambda \cdot S_\lambda = \cos^2\phi. \end{aligned}$$

Thus on the sphere

$$ds^2 = d\phi^2 + \cos^2\phi\, d\lambda^2, \qquad dA = \sqrt{EG - F^2}\, d\phi\, d\lambda = \cos\phi\, d\phi\, d\lambda.$$

The partial derivatives are

$$S_\phi = (-\sin\phi\cos\lambda,\ -\sin\phi\sin\lambda,\ \cos\phi), \qquad S_\lambda = (-\cos\phi\sin\lambda,\ \cos\phi\cos\lambda,\ 0).$$

We use the Euclidean dot product in $\mathbb{R}^3$: for $a = (a_1, a_2, a_3)$ and $b = (b_1, b_2, b_3)$,

$$a \cdot b = a_1b_1 + a_2b_2 + a_3b_3.$$

Coefficient E.

$$\begin{aligned}
E &= S_\phi \cdot S_\phi \\
&= (-\sin\phi\cos\lambda)^2 + (-\sin\phi\sin\lambda)^2 + (\cos\phi)^2 \\
&= \sin^2\phi\cos^2\lambda + \sin^2\phi\sin^2\lambda + \cos^2\phi \\
&= \sin^2\phi(\cos^2\lambda + \sin^2\lambda) + \cos^2\phi \\
&= \sin^2\phi + \cos^2\phi = 1.
\end{aligned}$$

Coefficient F.

$$\begin{aligned}
F &= S_\phi \cdot S_\lambda \\
&= (-\sin\phi\cos\lambda)(-\cos\phi\sin\lambda) + (-\sin\phi\sin\lambda)(\cos\phi\cos\lambda) + (\cos\phi)\cdot 0 \\
&= \sin\phi\cos\phi\cos\lambda\sin\lambda - \sin\phi\cos\phi\sin\lambda\cos\lambda \\
&= 0.
\end{aligned}$$

Coefficient G.

$$\begin{aligned}
G &= S_\lambda \cdot S_\lambda \\
&= (-\cos\phi\sin\lambda)^2 + (\cos\phi\cos\lambda)^2 + 0^2 \\
&= \cos^2\phi\sin^2\lambda + \cos^2\phi\cos^2\lambda \\
&= \cos^2\phi(\sin^2\lambda + \cos^2\lambda) \\
&= \cos^2\phi.
\end{aligned}$$

Thus the first fundamental form on the unit sphere in coordinates (ϕ, λ) has coefficients

$$E = 1, \qquad F = 0, \qquad G = \cos^2\phi.$$

Mercator projection (angle preserving)

The Mercator map (for radius 1) is

$$x = \lambda, \qquad y = u(\phi) = \operatorname{artanh}(\sin\phi) = \ln\tan\Big(\frac{\pi}{4} + \frac{\phi}{2}\Big).$$

The inverse map is

$$\lambda = x, \qquad \sin\phi = \tanh y, \qquad \cos\phi = \operatorname{sech} y,$$

and

$$\frac{d\phi}{dy} = \operatorname{sech} y.$$

Metric in (x, y). From

$$ds^2 = d\phi^2 + \cos^2\phi\, d\lambda^2$$

we substitute $\phi = \phi(y)$ and $\lambda = x$:

$$d\phi = \frac{d\phi}{dy}\,dy = \operatorname{sech} y\, dy, \qquad d\lambda = dx.$$

Therefore

$$\begin{aligned}
ds^2 &= d\phi^2 + \cos^2\phi\, d\lambda^2 \\
&= (\operatorname{sech} y)^2 dy^2 + (\operatorname{sech} y)^2 dx^2 \\
&= \operatorname{sech}^2 y\,(dx^2 + dy^2).
\end{aligned}$$

Thus the pulled-back metric is

$$I_{\text{Merc}} = \operatorname{sech}^2 y\,(dx^2 + dy^2),$$

a scalar multiple of the Euclidean metric: the map is conformal (angle preserving).

Area element. Starting from $dA = \cos\phi\, d\phi\, d\lambda$ and using

$$\cos\phi = \operatorname{sech} y, \qquad d\phi = \operatorname{sech} y\, dy, \qquad d\lambda = dx,$$

we obtain

$$dA_{\text{sphere}} = \operatorname{sech} y \cdot \operatorname{sech} y\, dy\, dx = \operatorname{sech}^2 y\, dx\, dy.$$

So the Euclidean area element $dx\,dy$ on the map corresponds to $\operatorname{sech}^2 y\, dx\, dy$ on the sphere; the factor depends on y, hence the Mercator projection is *not* area preserving.

Horizontal cylindrical projection (area preserving)

Consider the horizontal cylindrical projection from the sphere to the cylinder/plane:

$$x = \lambda, \qquad y = z = \sin\phi.$$

The inverse map is

$$\lambda = x, \qquad \phi = \arcsin y, \qquad \cos\phi = \sqrt{1-y^2},$$

and

$$d\phi = \frac{dy}{\cos\phi} = \frac{dy}{\sqrt{1-y^2}}.$$

Metric in (x, y). Again start from

$$ds^2 = d\phi^2 + \cos^2\phi\, d\lambda^2.$$

Substituting

$$d\phi = \frac{dy}{\cos\phi}, \qquad d\lambda = dx,$$

we get

$$\begin{aligned} ds^2 &= \frac{dy^2}{\cos^2\phi} + \cos^2\phi\, dx^2 \\ &= \frac{dy^2}{1-y^2} + (1-y^2)\, dx^2. \end{aligned}$$

This is not a scalar multiple of $dx^2 + dy^2$, so the projection is not conformal (angles are distorted).

Area element. For the area element,

$$dA = \cos\phi\, d\phi\, d\lambda,$$

we substitute

$$\cos\phi\, d\phi = \cos\phi \cdot \frac{dy}{\cos\phi} = dy, \qquad d\lambda = dx,$$

and obtain

$$dA_{\text{sphere}} = dy\, dx = dx\, dy.$$

Thus spherical area equals Euclidean area in (x, y): the horizontal cylindrical projection is *area preserving*, but not angle preserving.

Mercator vs. horizontal cylindrical projection

Both maps use a cylinder, but they are *not* the same projection.

1. Horizontal cylindrical (equal–area) projection

In geographic coordinates (ϕ, λ) (latitude, longitude), the *horizontal cylindrical projection* is given by

$$x = \lambda, \qquad y = z = \sin\phi.$$

Geometric description:

- Place the unit sphere inside a cylinder of radius 1.
- For each point (x, y, z) on the sphere, project it *horizontally* (radially from the z-axis) onto the cylinder: the height z is kept the same, and the horizontal coordinates are normalised to radius 1.
- Then unwrap the cylinder to the plane with coordinates (x, y).

Using the spherical metric

$$ds^2 = d\phi^2 + \cos^2\phi\, d\lambda^2, \qquad dA = \cos\phi\, d\phi\, d\lambda,$$

and the change of variables

$$x = \lambda, \qquad y = \sin\phi,$$

one finds

$$dA_{\text{sphere}} = dx\, dy.$$

Thus the horizontal cylindrical projection is *area preserving*, but it is not conformal: it does not preserve angles.

This projection is an example of a *Lambert cylindrical equal–area* projection (with standard parallel at the equator).

2. Mercator cylindrical (conformal) projection

The Mercator projection (for unit sphere) is given by

$$x = \lambda, \qquad y = u(\phi) = \operatorname{artanh}(\sin\phi) = \ln\tan\left(\frac{\pi}{4} + \frac{\phi}{2}\right).$$

Here the parallels (lines of constant latitude ϕ) are placed at heights $y(\phi)$ chosen so that the pulled-back metric on the map is

$$ds^2 = \operatorname{sech}^2 y\, (dx^2 + dy^2),$$

i.e. a scalar multiple of the Euclidean metric. Hence Mercator is a *conformal* projection: it preserves angles.

For the area element one obtains

$$dA_{\text{sphere}} = \operatorname{sech}^2 y\, dx\, dy,$$

so the factor depends on y, and the Mercator projection is *not* area preserving: regions at high latitudes appear greatly enlarged compared to regions near the equator.

3. Conclusion

Both projections are *cylindrical* in the sense that:

- longitudes $\lambda = \text{const}$ are mapped to vertical lines $x = \text{const}$,
- parallels $\phi = \text{const}$ are mapped to horizontal lines $y = \text{const}$,

but the rule for assigning the y–coordinate is different:

Horizontal cylindrical:	$y = \sin\phi$	(area preserving, not conformal),
Mercator:	$y = \operatorname{artanh}(\sin\phi)$	(conformal, not area preserving).

Therefore, the Mercator map is *not* the same as the horizontal projection onto a cylinder: they are two distinct cylindrical projections with different geometric properties.

Preliminaries for Problems 5–7

In this section we collect the basic definitions and formulas that will be used in Problems 5, 6 and 7.

Definition 8 (Parametrised surface and regularity)**.** *Let $U \subset \mathbb{R}^2$ be an open set. A map*

$$X : U \longrightarrow \mathbb{R}^3$$

is called a parametrised surface *if X is of class C^∞. We say that X is* regular *if for every $(u, v) \in U$ the partial derivatives $X_u(u, v)$ and $X_v(u, v)$ are linearly independent.*

Definition 9 (First fundamental form)**.** *Let $X : U \to \mathbb{R}^3$ be a regular parametrised surface. The* first fundamental form *at $(u, v) \in U$ is the quadratic form*

$$I = E\,du^2 + 2F\,du\,dv + G\,dv^2,$$

where

$$E = X_u \cdot X_u, \qquad F = X_u \cdot X_v, \qquad G = X_v \cdot X_v.$$

The associated area element is

$$dA = \sqrt{EG - F^2}\,du\,dv.$$

Definition 10 (Unit normal and Gauss map)**.** *For a regular parametrised surface $X : U \to \mathbb{R}^3$ the* unit normal vector field *is*

$$n(u, v) = \frac{X_u(u, v) \times X_v(u, v)}{\|X_u(u, v) \times X_v(u, v)\|}.$$

If $\Sigma = X(U)$ is the corresponding surface, the map

$$n : \Sigma \longrightarrow S^2, \qquad n\big(X(u, v)\big) = n(u, v),$$

is called the Gauss map *of Σ.*

Definition 11 (Second fundamental form)**.** *With X and n as above, the* second fundamental form *at $(u,v) \in U$ is the quadratic form*

$$II = L\,du^2 + 2M\,du\,dv + N\,dv^2,$$

where

$$L = X_{uu} \cdot n, \qquad M = X_{uv} \cdot n, \qquad N = X_{vv} \cdot n.$$

Definition 12 (Gauss curvature)**.** *The* Gauss curvature *of a regular parametrised surface $X : U \to \mathbb{R}^3$ is the function*

$$\kappa = \frac{LN - M^2}{EG - F^2},$$

where E, F, G and L, M, N are the coefficients of the first and second fundamental forms as above. In the special case $F = M = 0$ this reduces to

$$\kappa = \frac{L\,N}{E\,G}.$$

Definition 13 (Surface of revolution about the z-axis)**.** *Let $\eta : (a,b) \to \mathbb{R}^3$ be a smooth curve of the form*

$$\eta(t) = (f(t), 0, g(t)), \qquad f(t) > 0.$$

The surface of revolution *obtained by rotating η about the z-axis is the surface $\Sigma_\eta \subset \mathbb{R}^3$ parametrised by*

$$X : (a,b) \times (0, 2\pi) \longrightarrow \mathbb{R}^3, \qquad X(t,\theta) = \big(f(t)\cos\theta,\ f(t)\sin\theta,\ g(t)\big).$$

Definition 14 (Hyperboloids)**.** *The* hyperboloid of one sheet *is the surface*

$$H_1 = \{(x,y,z) \in \mathbb{R}^3 : x^2 + y^2 = z^2 + 1\}.$$

The hyperboloid of two sheets *is the surface*

$$H_2 = \{(x,y,z) \in \mathbb{R}^3 : x^2 + y^2 = z^2 - 1,\ |z| > 1\}.$$

Both H_1 and H_2 can be realised as surfaces of revolution by taking suitable profile curves $\eta(t) = (f(t), 0, g(t))$ in the xz-plane and rotating them about the z-axis.

Definition 15 (Standard torus of revolution)**.** *Let $R > r > 0$. The* standard torus of revolution *with major radius R and minor radius r is the surface*

$$T_{R,r} = \left\{\, \big((R + r\cos u)\cos v,\ (R + r\cos u)\sin v,\ r\sin u\big) : 0 < u, v < 2\pi \,\right\}.$$

In Problem 7 we take $R = 2$ and $r = 1$, so the torus $T \subset \mathbb{R}^3$ is parametrised by

$$X(u,v) = \big((2 + \cos u)\cos v,\ (2 + \cos u)\sin v,\ \sin u\big), \qquad 0 < u, v < 2\pi.$$

Definition 16 (Rotation about the z-axis)**.** *For $\theta \in \mathbb{R}$, the* rotation of angle θ about the z-axis *is the linear isometry $R_\theta : \mathbb{R}^3 \to \mathbb{R}^3$ given by*

$$R_\theta(x,y,z) = (x\cos\theta - y\sin\theta,\ x\sin\theta + y\cos\theta,\ z).$$

In particular R_θ preserves any surface of revolution about the z-axis, and we will use this in Problem 6.

Geometric differences between the two hyperboloids

We use cylindrical coordinates (r, θ, z) with

$$r = \sqrt{x^2 + y^2}.$$

Hyperboloid of one sheet

$$x^2 + y^2 = z^2 + 1 \quad \iff \quad r^2 = z^2 + 1.$$

Profile in the (r, z)–plane.

$$r(z) = \sqrt{1 + z^2}, \qquad z \in \mathbb{R}.$$

This curve is defined for all z and is connected; it passes near the origin and widens as $|z|$ increases.

3D shape. The surface is a single connected "hourglass/waist" shape. For each plane $z = \text{const}$, the intersection with the surface is a circle.

Asymptotics. As $|z| \to \infty$,

$$r(z) \sim |z|,$$

so the surface is asymptotic to the cone $r = |z|$.

Gauss curvature. For the hyperboloid of one sheet the Gauss curvature is

$$\kappa_1(z) = -\frac{1}{(1 + 2z^2)^2} < 0 \qquad \text{for all } z \in \mathbb{R}.$$

Thus the surface is everywhere saddle-shaped (negative curvature), and $\kappa_1(z) \to 0$ as $|z| \to \infty$.

Hyperboloid of two sheets

$$x^2 + y^2 = z^2 - 1 \quad \iff \quad r^2 = z^2 - 1.$$

Profile in the (r, z)–plane.

$$r(z) = \sqrt{z^2 - 1}, \qquad |z| > 1.$$

The profile is defined only for $|z| > 1$. There is a gap between the two branches for $|z| \leq 1$.

3D shape. The surface consists of two disconnected "bowls": one above and one below the origin. For $|z| > 1$, each plane $z = \text{const}$ intersects the surface in a circle; for $|z| \leq 1$ there is no intersection.

Asymptotics. Again,

$$r(z) \sim |z| \qquad \text{as } |z| \to \infty,$$

so the two-sheet hyperboloid is also asymptotic to the cone $r = |z|$.

Gauss curvature. For the hyperboloid of two sheets the Gauss curvature is

$$\kappa_2(z) = \frac{1}{(2z^2-1)^2} > 0 \qquad \text{for } |z| > 1.$$

Each sheet is locally convex (positive curvatur

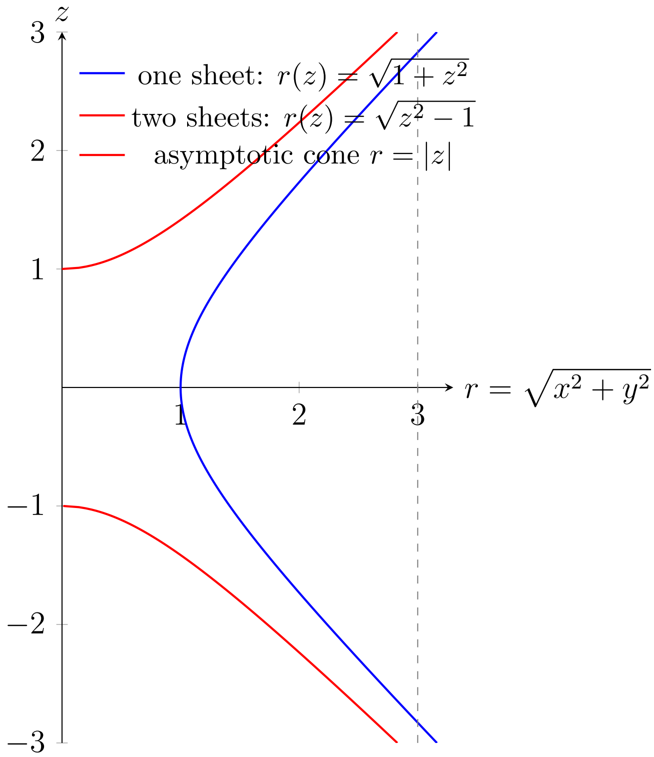

Figure 1: Profiles of the hyperboloid of one sheet (blue) and two sheets (red) in the (r, z)-plane. The dashed line is the common asymptotic cone $r = |z|$. The one-sheet surface exists for all z, while the two-sheet surface exists only for $|z| > 1$, leaving a gap between the sheets.

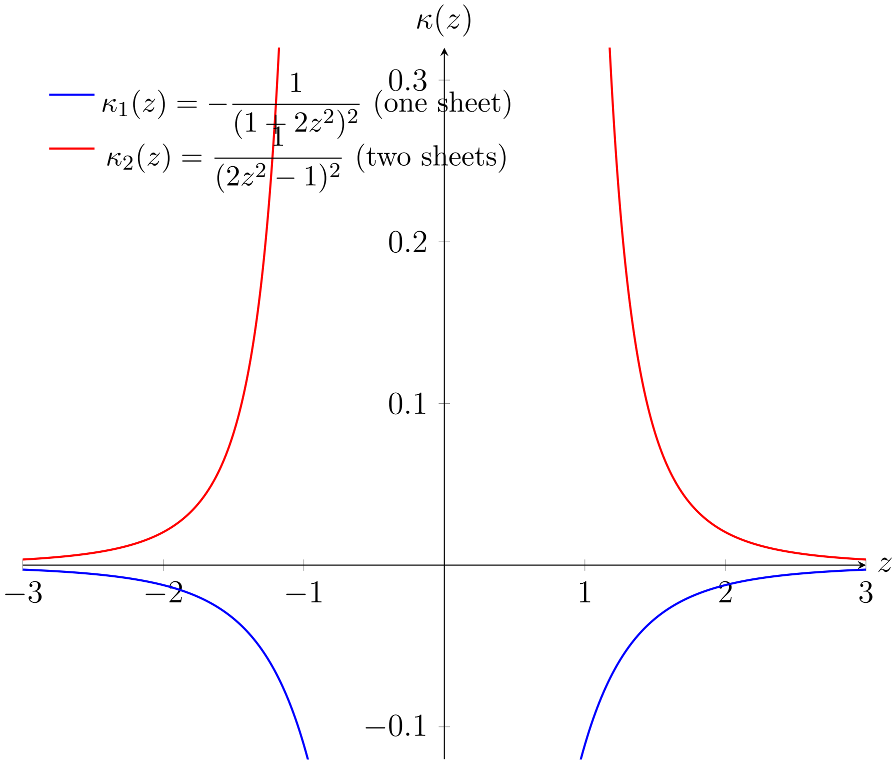

Figure 2: Gauss curvature comparison. For the hyperboloid of one sheet (blue), $\kappa_1(z) < 0$ for all z (saddle surface). For the hyperboloid of two sheets (red), $\kappa_2(z) > 0$ for $|z| > 1$ (locally convex sheets). In both cases $\kappa(z) \to 0$ as $|z| \to \infty$.

Problem 5: Gauss curvature of a surface of revolution

Let $\eta : (a,b) \to \mathbb{R}^3$ be a smooth curve

$$\eta(t) = (f(t), 0, g(t)), \qquad f(t) > 0, \quad \eta'(t) \neq 0,$$

which is a homeomorphism onto its image. Let Σ_η be the surface of revolution obtained by rotating η about the z-axis.

(a) General formula for κ and the arc-length case

We parametrise Σ_η by

$$X : (a,b) \times (0, 2\pi) \longrightarrow \mathbb{R}^3, \qquad X(t,\theta) = \big(f(t)\cos\theta,\ f(t)\sin\theta,\ g(t)\big).$$

First derivatives.

$$\begin{aligned} X_t &= (f'(t)\cos\theta,\ f'(t)\sin\theta,\ g'(t)), \\ X_\theta &= (-f(t)\sin\theta,\ f(t)\cos\theta,\ 0). \end{aligned}$$

First fundamental form.

$$\begin{aligned} E &= X_t \cdot X_t = (f')^2(\cos^2\theta + \sin^2\theta) + (g')^2 = (f')^2 + (g')^2, \\ F &= X_t \cdot X_\theta = f'f(-\cos\theta\sin\theta + \sin\theta\cos\theta) = 0, \\ G &= X_\theta \cdot X_\theta = f^2(\sin^2\theta + \cos^2\theta) = f^2. \end{aligned}$$

Normal vector. The (non–unit) normal is

$$X_t \times X_\theta = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ f'\cos\theta & f'\sin\theta & g' \\ -f\sin\theta & f\cos\theta & 0 \end{vmatrix} = \big(-ff'\cos\theta,\ -ff'\sin\theta,\ -fg'\big).$$

Its length is

$$\|X_t \times X_\theta\| = f\sqrt{(f')^2 + (g')^2} = f\sqrt{E}.$$

Thus we can choose the unit normal

$$n(t,\theta) = \frac{X_t \times X_\theta}{\|X_t \times X_\theta\|} = \frac{1}{\sqrt{(f')^2 + (g')^2}}\big(-g'\cos\theta,\ -g'\sin\theta,\ f'\big).$$

Second derivatives.

$$\begin{aligned} X_{tt} &= (f''\cos\theta,\ f''\sin\theta,\ g''), \\ X_{t\theta} &= (-f'\sin\theta,\ f'\cos\theta,\ 0), \\ X_{\theta\theta} &= (-f\cos\theta,\ -f\sin\theta,\ 0). \end{aligned}$$

Second fundamental form. We compute

$$L = X_{tt} \cdot n, \qquad M = X_{t\theta} \cdot n, \qquad N = X_{\theta\theta} \cdot n.$$

First,

$$\begin{aligned} L &= X_{tt} \cdot n \\ &= \frac{1}{\sqrt{(f')^2 + (g')^2}} \Big[f'' \cos\theta(-g' \cos\theta) + f'' \sin\theta(-g' \sin\theta) + g'' f' \Big] \\ &= \frac{1}{\sqrt{(f')^2 + (g')^2}} \Big[-f'' g' (\cos^2\theta + \sin^2\theta) + g'' f' \Big] \\ &= \frac{f' g'' - f'' g'}{\sqrt{(f')^2 + (g')^2}}. \end{aligned}$$

Next,

$$\begin{aligned} M &= X_{t\theta} \cdot n \\ &= \frac{1}{\sqrt{(f')^2 + (g')^2}} \Big[(-f' \sin\theta)(-g' \cos\theta) + (f' \cos\theta)(-g' \sin\theta) + 0 \cdot f' \Big] \\ &= \frac{f' g' (\sin\theta\cos\theta - \cos\theta\sin\theta)}{\sqrt{(f')^2 + (g')^2}} = 0. \end{aligned}$$

Finally,

$$\begin{aligned} N &= X_{\theta\theta} \cdot n \\ &= \frac{1}{\sqrt{(f')^2 + (g')^2}} \Big[(-f \cos\theta)(-g' \cos\theta) + (-f \sin\theta)(-g' \sin\theta) + 0 \cdot f' \Big] \\ &= \frac{f g' (\cos^2\theta + \sin^2\theta)}{\sqrt{(f')^2 + (g')^2}} = \frac{f g'}{\sqrt{(f')^2 + (g')^2}}. \end{aligned}$$

Gauss curvature. Since $F = M = 0$, the Gauss curvature is

$$\kappa = \frac{L\,N}{E\,G}.$$

Thus

$$\begin{aligned} \kappa &= \frac{\dfrac{f' g'' - f'' g'}{\sqrt{(f')^2 + (g')^2}} \cdot \dfrac{f g'}{\sqrt{(f')^2 + (g')^2}}}{((f')^2 + (g')^2)\, f^2} \\ &= \frac{(f' g'' - f'' g')\, f g'}{((f')^2 + (g')^2)^2\, f^2} \\ &= \frac{(f' g'' - f'' g')\, g'}{((f')^2 + (g')^2)^2\, f}, \end{aligned}$$

which is the desired formula.

Arc-length parametrisation. Assume that t is arc-length, so

$$(f')^2 + (g')^2 \equiv 1.$$

Differentiating,

$$2f'f'' + 2g'g'' = 0 \quad \Longrightarrow \quad f'f'' + g'g'' = 0 \quad \Longrightarrow \quad g'' = -\frac{f'f''}{g'}.$$

Substitute this in the numerator of κ:

$$\begin{aligned}(f'g'' - f''g')\, g' &= \Big(f'\Big(-\frac{f'f''}{g'}\Big) - f''g'\Big)g' \\ &= \Big(-\frac{(f')^2 f''}{g'} - f''g'\Big)g' \\ &= -(f')^2 f'' - f''(g')^2 = -f''\big((f')^2 + (g')^2\big) = -f''.\end{aligned}$$

Since $E = (f')^2 + (g')^2 \equiv 1$, we obtain

$$\kappa = \frac{-f''}{1^2\, f} = -\frac{f''}{f}.$$

(b) Hyperboloids

Hyperboloid of one sheet: $x^2 + y^2 = z^2 + 1$. We set $t = z$ and write the meridian curve as

$$\eta(t) = (f(t), 0, g(t)), \qquad f(t) = \sqrt{1+t^2}, \quad g(t) = t.$$

Then

$$f'(t) = \frac{t}{\sqrt{1+t^2}}, \qquad f''(t) = \frac{1}{(1+t^2)^{3/2}}, \qquad g'(t) = 1, \quad g''(t) = 0.$$

The general formula for κ gives

$$\kappa(t) = \frac{(f'g'' - f''g')\, g'}{\big((f')^2 + (g')^2\big)^2 f} = \frac{-f''}{\big((f')^2 + 1\big)^2 f}.$$

Now

$$(f')^2 + 1 = \frac{t^2}{1+t^2} + 1 = \frac{1+2t^2}{1+t^2}, \qquad f(t) = \sqrt{1+t^2}.$$

Hence

$$\begin{aligned}\kappa(t) &= -\frac{1}{(1+t^2)^{3/2}} \cdot \frac{(1+t^2)^2}{(1+2t^2)^2} \cdot \frac{1}{\sqrt{1+t^2}} \\ &= -\frac{1}{(1+2t^2)^2}.\end{aligned}$$

Therefore $\kappa(t) < 0$ for all t and

$$\kappa(t) \longrightarrow 0^- \qquad \text{as } |t| \to \infty.$$

Hyperboloid of two sheets: $x^2 + y^2 = z^2 - 1$. For the upper sheet, take $t = z > 1$ and

$$\eta(t) = (f(t), 0, g(t)), \qquad f(t) = \sqrt{t^2 - 1}, \quad g(t) = t.$$

Then

$$f'(t) = \frac{t}{\sqrt{t^2 - 1}}, \qquad f''(t) = -\frac{t^2}{(t^2 - 1)^{3/2}} + \frac{1}{\sqrt{t^2 - 1}}, \qquad g'(t) = 1, \quad g''(t) = 0.$$

Again

$$\kappa(t) = \frac{-f''}{\left((f')^2 + 1\right)^2 f}.$$

We compute

$$(f')^2 + 1 = \frac{t^2}{t^2 - 1} + 1 = \frac{2t^2 - 1}{t^2 - 1}, \qquad f(t) = \sqrt{t^2 - 1}.$$

A straightforward simplification yields

$$\kappa(t) = \frac{1}{(2t^2 - 1)^2}, \qquad t > 1.$$

Thus $\kappa(t) > 0$ everywhere on each sheet and

$$\kappa(t) \longrightarrow 0^+ \qquad \text{as } t \to \infty.$$

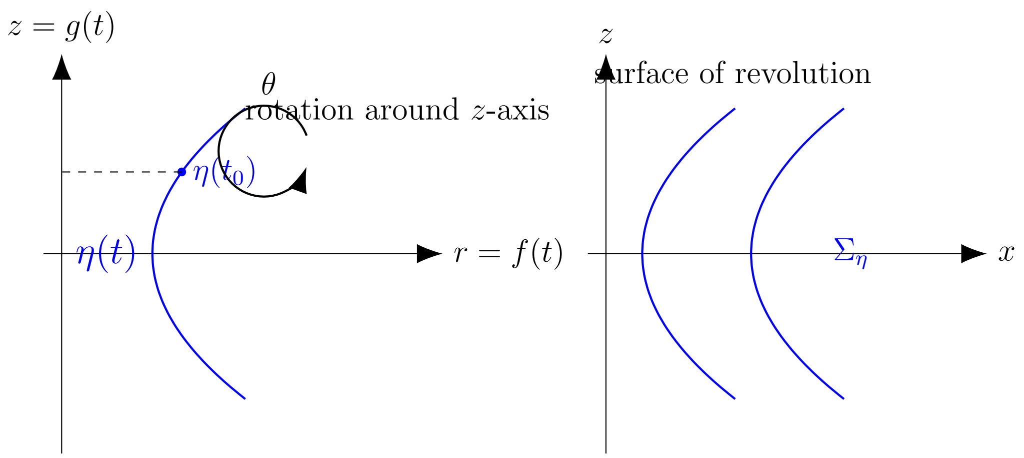

Figure 3: Profile curve $\eta(t) = (f(t), 0, g(t))$ in the meridian plane and the associated surface of revolution Σ_η obtained by rotating around the z-axis.

Problem 6: Gauss map and rotations

Problem 6

Let Σ_η be as in the previous question. Let

$$n : \Sigma_\eta \to S^2$$

be the Gauss map. Let $R_\theta : \mathbb{R}^3 \to \mathbb{R}^3$ denote rotation by angle θ about the z-axis.

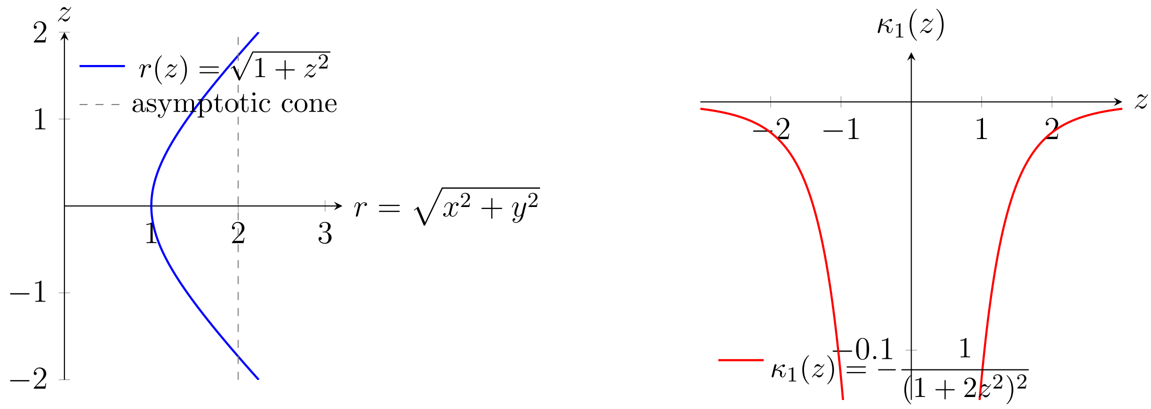

Figure 4: Hyperboloid of one sheet $x^2 + y^2 = z^2 + 1$. Left: profile curve in the (r, z)-plane, with dashed asymptotic cone. Right: Gauss curvature $\kappa_1(z)$, negative for all z and tending to 0 as $|z| \to \infty$.

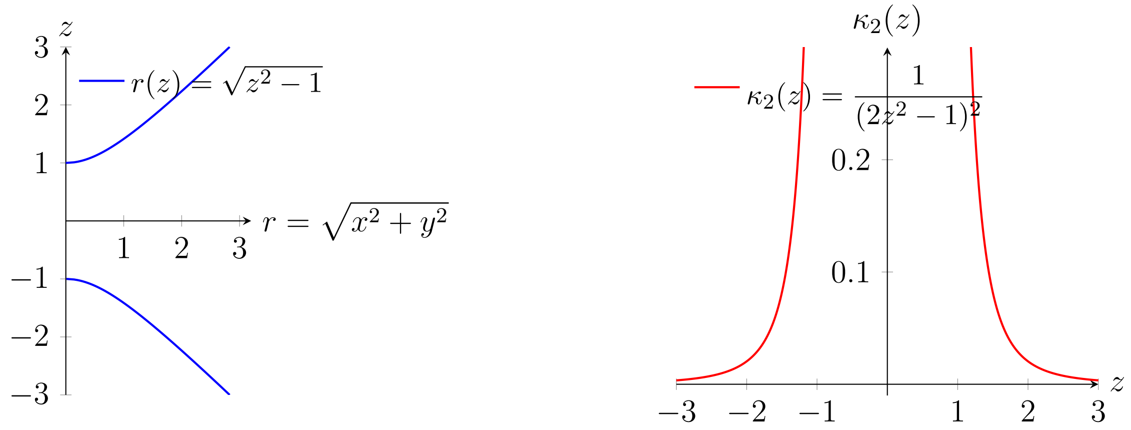

Figure 5: Hyperboloid of two sheets $x^2 + y^2 = z^2 - 1$. Left: profile of the upper and lower sheets in the (r, z)-plane. Right: Gauss curvature $\kappa_2(z)$, positive for $|z| > 1$ and decreasing to 0 as $|z| \to \infty$.

(c) Prove that for $p \in \Sigma_\eta$,
$$n(R_\theta(p)) = R_\theta(n(p)).$$

(d) Hence, or otherwise, prove that for the hyperboloid
$$\{x^2 + y^2 = z^2 + 1\},$$
the image of the Gauss map is the open annulus
$$\{\, |z| < 1/\sqrt{2}\,\} \subset S^2.$$

Gauss map and its basic properties

Let $\Sigma \subset \mathbb{R}^3$ be an oriented smooth surface.

Definition 17 (Unit normal and Gauss map)**.** *For each $p \in \Sigma$ there exists a unique unit normal vector $n(p)$ compatible with the orientation, i.e.*
$$n(p) \perp T_p\Sigma, \qquad \|n(p)\| = 1.$$

The map

$$n : \Sigma \longrightarrow S^2, \qquad p \longmapsto n(p),$$

is called the Gauss map *of* Σ.

Example 0.1 (Sphere)**.** For the unit sphere $S^2 = \{x \in \mathbb{R}^3 : \|x\| = 1\}$ with outward orientation, the normal at p is $n(p) = p$. Thus the Gauss map $n : S^2 \to S^2$ is the identity map.

Example 0.2 (Plane)**.** For an oriented plane $ax + by + cz + d = 0$ the unit normal is constant, so the Gauss map is constant and its image is a single point of S^2.

Example 0.3 (Surface of revolution)**.** Let

$$X(t, \theta) = \big(f(t)\cos\theta,\ f(t)\sin\theta,\ g(t)\big), \qquad f(t) > 0,$$

parametrise a surface of revolution about the z-axis. Then a unit normal is

$$n(t, \theta) = \frac{1}{\sqrt{(f'(t))^2 + (g'(t))^2}}\big(-g'(t)\cos\theta,\ -g'(t)\sin\theta,\ f'(t)\big),$$

and this defines the Gauss map $n : \Sigma_\eta \to S^2$. For rotation R_θ about the z-axis one has the equivariance

$$n(R_\theta(p)) = R_\theta\big(n(p)\big), \qquad p \in \Sigma_\eta.$$

Definition 18 (Shape operator and curvature via Gauss map)**.** *At each point $p \in \Sigma$, the differential of the Gauss map*

$$dn_p : T_p\Sigma \longrightarrow T_{n(p)}S^2$$

is (up to sign) the shape operator S_p. *Its eigenvalues $k_1(p), k_2(p)$ are the principal curvatures at p. The Gauss and mean curvatures can then be expressed as*

$$K(p) = \det(dn_p) = k_1(p)k_2(p), \qquad H(p) = \tfrac{1}{2}\operatorname{tr}(dn_p) = \tfrac{1}{2}\big(k_1(p) + k_2(p)\big).$$

Remark 0.4. The image $n(\Sigma) \subset S^2$ encodes global information about the surface. For instance, for the hyperboloid of one sheet $x^2 + y^2 = z^2 + 1$ the Gauss map image is the open spherical annulus

$$n(\Sigma) = \{\, q \in S^2 : |q_3| < 1/\sqrt{2}\,\},$$

while for a plane it is a single point and for the unit sphere it is all of S^2.

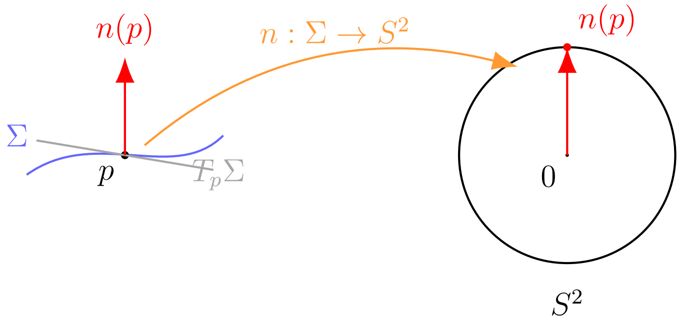

Figure 6: The Gauss map assigns to each point $p \in \Sigma$ its unit normal vector $n(p)$, viewed as a point on the unit sphere S^2.

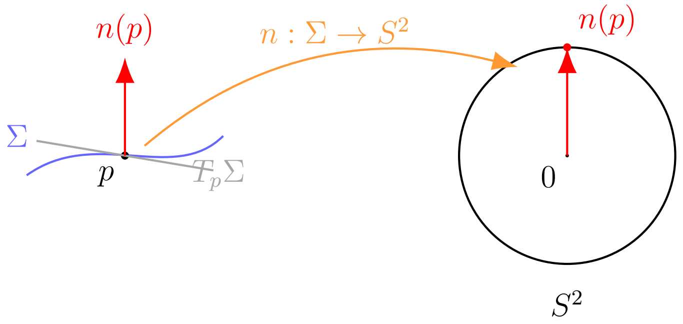

Figure 7: The Gauss map assigns to each point $p \in \Sigma$ its unit normal vector $n(p)$, viewed as a point on the unit sphere S^2.

Let Σ_η be as in Problem 5 and

$$n : \Sigma_\eta \longrightarrow S^2$$

its Gauss map. Let $R_\theta : \mathbb{R}^3 \to \mathbb{R}^3$ denote rotation by angle θ about the z-axis.

(c) Commutation of Gauss map with rotation

We must show that for every $p \in \Sigma_\eta$,

$$n(R_\theta(p)) = R_\theta\big(n(p)\big).$$

Proof. First note that R_θ preserves Σ_η because Σ_η is a surface of revolution with respect to the z-axis.

Let $p \in \Sigma_\eta$ and write $A = R_\theta$ for brevity. Since A is an isometry of $\mathbb{R}^3$, its differential $dA_p : T_p\mathbb{R}^3 \to T_{A(p)}\mathbb{R}^3$ is an orthogonal transformation. Because $A(\Sigma_\eta) = \Sigma_\eta$, we have

$$dA_p\big(T_p\Sigma_\eta\big) = T_{A(p)}\Sigma_\eta.$$

The unit normal $n(p)$ is characterised by

$$n(p) \perp T_p\Sigma_\eta, \qquad \|n(p)\| = 1.$$

Orthogonality of dA_p implies that $dA_p(n(p))$ is orthogonal to $dA_p(T_p\Sigma_\eta) = T_{A(p)}\Sigma_\eta$ and has length 1. Hence $dA_p(n(p))$ is a unit normal at $A(p)$, so it must equal $n(A(p))$ up to sign. Since A preserves orientation, the sign is $+$ and

$$n(A(p)) = dA_p(n(p)).$$

But for a rotation about the z-axis, dA_p is just the linear map R_θ itself, so

$$n(R_\theta(p)) = R_\theta(n(p)),$$

as required.

Equivalently, one can check this explicitly using the parametrisation $X(t,\varphi)$:

$$R_\theta(X(t,\varphi)) = X(t,\varphi+\theta),$$

and the explicit formula

$$n(t,\varphi) = \frac{1}{\sqrt{(f')^2+(g')^2}}\big(-g'\cos\varphi,\ -g'\sin\varphi,\ f'\big)$$

satisfies

$$n(t,\varphi+\theta) = R_\theta\big(n(t,\varphi)\big).$$

(d) Image of the Gauss map for $x^2+y^2=z^2+1$

Consider the hyperboloid of one sheet

$$\Sigma = \{(x,y,z) \in \mathbb{R}^3 : x^2+y^2 = z^2+1\}.$$

We use the parametrisation from Problem 5(b):

$$X(t,\theta) = \big(\sqrt{1+t^2}\cos\theta,\ \sqrt{1+t^2}\sin\theta,\ t\big), \qquad t \in \mathbb{R},\ \theta \in (0,2\pi).$$

Normal vector. From the general formula,

$$n(t,\theta) = \frac{1}{\sqrt{(f')^2+1}}\big(-\cos\theta,\ -\sin\theta,\ f'(t)\big)$$

because here $g'(t) = 1$ and $f'(t) = t/\sqrt{1+t^2}$. Indeed,

$$(f')^2 + 1 = \frac{t^2}{1+t^2} + 1 = \frac{1+2t^2}{1+t^2}.$$

Thus the third component of n is

$$n_3(t) = \frac{f'(t)}{\sqrt{(f')^2+1}} = \frac{\dfrac{t}{\sqrt{1+t^2}}}{\sqrt{\dfrac{1+2t^2}{1+t^2}}} = \frac{t}{\sqrt{1+2t^2}}.$$

Range of n_3. For all $t \in \mathbb{R}$,

$$|n_3(t)| = \frac{|t|}{\sqrt{1+2t^2}} < \frac{|t|}{\sqrt{2t^2}} = \frac{1}{\sqrt{2}}.$$

Moreover,

$$\lim_{t\to\pm\infty} n_3(t) = \pm\frac{1}{\sqrt{2}}.$$

The function $n_3 : \mathbb{R} \to (-1/\sqrt{2}, 1/\sqrt{2})$ is continuous and strictly increasing, so its image is the entire open interval $(-1/\sqrt{2}, 1/\sqrt{2})$.

Image of the Gauss map. By part (c), the Gauss map is equivariant with respect to rotations about the z-axis: for each θ,

$$n(t,\theta) = R_\theta\big(n(t,0)\big).$$

Thus, for each fixed t, as θ varies, the image of $n(t,\theta)$ is the full latitude circle on S^2 at height $n_3(t)$.

Since $n_3(t)$ ranges over $(-1/\sqrt{2}, 1/\sqrt{2})$, the image of the Gauss map is exactly the open spherical annulus

$$\{\, q \in S^2 : |q_3| < 1/\sqrt{2}\,\},$$

where q_3 denotes the z-coordinate of q.

Problem 7: Gauss curvature on a torus of revolution

Problem 7

Let $T \subset \mathbb{R}^3$ be the smooth embedded torus obtained by rotating the circle

$$(x-2)^2 + z^2 = 1$$

in the xz-plane around the z-axis. Sketch T, and draw an illustration of where on T the Gauss curvature κ is positive, negative and zero.

Let $T \subset \mathbb{R}^3$ be the torus obtained by rotating the circle

$$(x-2)^2 + z^2 = 1$$

in the xz-plane about the z-axis. This is a torus of revolution with major radius $R = 2$ and minor radius $r = 1$.

We parametrise T by

$$X(u,v) = \big((2+\cos u)\cos v,\ (2+\cos u)\sin v,\ \sin u\big), \qquad 0 < u, v < 2\pi.$$

Computation of the Gauss curvature

First derivatives.

$$\begin{aligned} X_u &= (-\sin u \cos v,\ -\sin u \sin v,\ \cos u), \\ X_v &= (-(2+\cos u)\sin v,\ (2+\cos u)\cos v,\ 0). \end{aligned}$$

First fundamental form.

$$\begin{aligned} E &= X_u \cdot X_u = \sin^2 u(\cos^2 v + \sin^2 v) + \cos^2 u = 1, \\ F &= X_u \cdot X_v \\ &= (-\sin u \cos v)(-(2+\cos u)\sin v) + (-\sin u \sin v)((2+\cos u)\cos v) + \cos u \cdot 0 \\ &= \sin u(2+\cos u)(\cos v \sin v - \sin v \cos v) = 0, \\ G &= X_v \cdot X_v = (2+\cos u)^2(\sin^2 v + \cos^2 v) = (2+\cos u)^2. \end{aligned}$$

Normal vector. The cross product is

$$\begin{aligned} X_u \times X_v &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -\sin u \cos v & -\sin u \sin v & \cos u \\ -(2+\cos u)\sin v & (2+\cos u)\cos v & 0 \end{vmatrix} \\ &= \big(-(2+\cos u)\cos u \cos v,\ -(2+\cos u)\cos u \sin v,\ -(2+\cos u)\sin u\big). \end{aligned}$$

Its length is

$$\|X_u \times X_v\| = |2+\cos u|\sqrt{\cos^2 u(\cos^2 v + \sin^2 v) + \sin^2 u} = 2 + \cos u,$$

since $2 + \cos u > 0$. We take

$$n(u,v) = \frac{X_u \times X_v}{2+\cos u} = (-\cos u \cos v,\ -\cos u \sin v,\ -\sin u).$$

Second derivatives.

$$\begin{aligned} X_{uu} &= (-\cos u \cos v,\ -\cos u \sin v,\ -\sin u), \\ X_{uv} &= (\sin u \sin v,\ -\sin u \cos v,\ 0), \\ X_{vv} &= (-(2+\cos u)\cos v,\ -(2+\cos u)\sin v,\ 0). \end{aligned}$$

Second fundamental form. We compute

$$L = X_{uu} \cdot n, \qquad M = X_{uv} \cdot n, \qquad N = X_{vv} \cdot n.$$

First,

$$\begin{aligned} L &= X_{uu} \cdot n \\ &= (-\cos u \cos v)(-\cos u \cos v) + (-\cos u \sin v)(-\cos u \sin v) + (-\sin u)(-\sin u) \\ &= \cos^2 u(\cos^2 v + \sin^2 v) + \sin^2 u = \cos^2 u + \sin^2 u = 1. \end{aligned}$$

Since $E = 1$ and $\|X_u \times X_v\| = 2 + \cos u$, the usual convention for the second fundamental coefficient is

$$L = \frac{X_{uu} \cdot (X_u \times X_v)}{\|X_u \times X_v\|} = \frac{1 \cdot (2 + \cos u)}{2 + \cos u} = \frac{1}{2 + \cos u}.$$

(Equivalently, one can divide by $2 + \cos u$ at the end.)

Next,

$$\begin{aligned} M &= X_{uv} \cdot n \\ &= (\sin u \sin v)(-\cos u \cos v) + (-\sin u \cos v)(-\cos u \sin v) + 0 \cdot (-\sin u) \\ &= -\sin u \cos u \sin v \cos v + \sin u \cos u \cos v \sin v = 0. \end{aligned}$$

Finally,

$$\begin{aligned} N &= X_{vv} \cdot n \\ &= (-(2 + \cos u) \cos v)(-\cos u \cos v) + (-(2 + \cos u) \sin v)(-\cos u \sin v) + 0 \cdot (-\sin u) \\ &= (2 + \cos u) \cos u(\cos^2 v + \sin^2 v) = (2 + \cos u) \cos u. \end{aligned}$$

Dividing by $\|X_u \times X_v\| = 2 + \cos u$, we obtain

$$N = \cos u.$$

Gauss curvature. Since $F = M = 0$,

$$\kappa(u, v) = \frac{L\,N}{E\,G} = \frac{\dfrac{1}{2 + \cos u} \cdot \cos u}{1 \cdot (2 + \cos u)^2} = \frac{\cos u}{(2 + \cos u)^3}.$$

Sign of the Gauss curvature

The denominator $(2 + \cos u)^3$ is strictly positive for all u, so the sign of κ is the sign of $\cos u$:

- If $\cos u > 0$ (outer part of the torus), then $\kappa > 0$: the surface is locally convex there.
- If $\cos u < 0$ (inner part of the torus near the hole), then $\kappa < 0$: the surface is saddle-shaped.
- If $\cos u = 0$ (i.e. $u = \frac{\pi}{2}$ or $u = \frac{3\pi}{2}$), then $\kappa = 0$: there are two parallel circles on the "top" and "bottom" of the torus where the Gauss curvature vanishes.

These circles separate the regions of positive and negative Gaussian curvature on the torus.

Problem 8 – A ruled surface with negative Gauss curvature

Problem. Consider the surface $\Sigma \subset \mathbb{R}^3$ with parametrization

$$\sigma(u,v) = \gamma(u) + v\,a(u), \qquad u \in [0,2\pi),\ v \in (-1,1),$$

where

$$\gamma(u) = (\cos u,\ \sin u,\ 0), \qquad a(u) = \big(\cos\tfrac{u}{2}\cos u,\ \cos\tfrac{u}{2}\sin u,\ \sin\tfrac{u}{2}\big).$$

(This is an example of a *ruled surface*: one which is locally swept out by a moving Euclidean straight line.)

1. Sketch Σ.
2. Prove that Σ is a smooth surface.
3. Show that the Gauss curvature κ of Σ is everywhere negative.

Theory. A parametrized surface $\sigma(u,v)$ is *regular* if $\sigma_u \times \sigma_v \neq 0$ everywhere. The first fundamental form has coefficients

$$E = \sigma_u \cdot \sigma_u, \quad F = \sigma_u \cdot \sigma_v, \quad G = \sigma_v \cdot \sigma_v.$$

The second fundamental form has coefficients

$$e = \sigma_{uu} \cdot n, \quad f = \sigma_{uv} \cdot n, \quad g = \sigma_{vv} \cdot n,$$

where $n = \dfrac{\sigma_u \times \sigma_v}{|\sigma_u \times \sigma_v|}$ is the unit normal. The Gauss curvature is

$$K = \frac{eg - f^2}{EG - F^2}.$$

For convenience one may use triple products:

$$e = \frac{\sigma_{uu} \cdot (\sigma_u \times \sigma_v)}{|\sigma_u \times \sigma_v|}, \quad f = \frac{\sigma_{uv} \cdot (\sigma_u \times \sigma_v)}{|\sigma_u \times \sigma_v|}, \quad g = \frac{\sigma_{vv} \cdot (\sigma_u \times \sigma_v)}{|\sigma_u \times \sigma_v|}.$$

Solution.

Geometry. The base curve $\gamma(u) = (\cos u, \sin u, 0)$ is the unit circle in the xy-plane. For each fixed u, the map $v \mapsto \sigma(u,v) = \gamma(u) + va(u)$ is a straight line through $\gamma(u)$ with direction $a(u)$. The vector $a(u)$ has length 1, so Σ is a ruled surface obtained by moving a unit line along the circle. Geometrically, Σ looks like a “saddle-shaped ring” wrapped around the circle.

Regularity. We have

$$\sigma_u = \gamma'(u) + va'(u), \qquad \sigma_v = a(u),$$

with

$$\gamma'(u) = (-\sin u, \cos u, 0).$$

A straightforward computation gives that

$$\big(\sigma_u \times \sigma_v\big)_3 = -(v\cos\tfrac{u}{2} + 1)\cos\tfrac{u}{2}.$$

For $u \neq \pi$ we have $\cos\frac{u}{2} \neq 0$, and since $|v| < 1$, the expression $v\cos\frac{u}{2} + 1$ cannot vanish. Thus the third component is non-zero.

For $u = \pi$, one checks explicitly that

$$\sigma_u \times \sigma_v\big|_{u=\pi} = (-1, -\tfrac{v}{2}, 0) \neq 0.$$

Therefore $\sigma_u \times \sigma_v \neq 0$ for all (u, v) in the domain, and σ is a regular parametrization. Hence Σ is a smooth surface.

First fundamental form. Because $|a(u)| \equiv 1$, differentiating $a \cdot a = 1$ yields $a'(u) \cdot a(u) = 0$. A direct computation also gives

$$\gamma'(u) \cdot a(u) = 0.$$

Therefore

$$F = \sigma_u \cdot \sigma_v = (\gamma'(u) + va'(u)) \cdot a(u) = \gamma'(u) \cdot a(u) + va'(u) \cdot a(u) = 0.$$

Next,

$$E = \sigma_u \cdot \sigma_u = |\gamma'(u) + va'(u)|^2 = |\gamma'(u)|^2 + 2v\,\gamma'(u) \cdot a'(u) + v^2|a'(u)|^2.$$

One finds

$$|\gamma'(u)|^2 = 1, \qquad \gamma'(u) \cdot a'(u) = \cos\tfrac{u}{2}, \qquad |a'(u)|^2 = \frac{\cos u}{2} + \frac{3}{4},$$

so

$$E(u, v) = 1 + 2v\cos\tfrac{u}{2} + v^2\left(\frac{\cos u}{2} + \frac{3}{4}\right) > 0.$$

Finally,

$$G = \sigma_v \cdot \sigma_v = a(u) \cdot a(u) = 1.$$

Thus the first fundamental form is

$$I = E\,du^2 + G\,dv^2, \qquad E > 0,\ G = 1,\ F = 0.$$

Second fundamental form and Gauss curvature. Second derivatives are

$$\sigma_{uu} = \gamma''(u) + va''(u), \qquad \sigma_{uv} = a'(u), \qquad \sigma_{vv} = 0.$$

Hence $g = \sigma_{vv} \cdot n = 0$.

Let $N = \sigma_u \times \sigma_v$. Then

$$f = \frac{\sigma_{uv} \cdot N}{|N|} = \frac{a'(u) \cdot N}{|N|}.$$

Since $N = (\gamma'(u) + va'(u)) \times a(u)$, we obtain

$$a'(u) \cdot N = a'(u) \cdot \big(\gamma'(u) \times a(u)\big) + v\, a'(u) \cdot \big(a'(u) \times a(u)\big) = a'(u) \cdot \big(\gamma'(u) \times a(u)\big),$$

because $x \cdot (x \times y) = 0$.

A direct computation shows

$$\gamma'(u) \times a(u) = \big(\sin \tfrac{u}{2} \cos u,\ \sin \tfrac{u}{2} \sin u,\ -\cos \tfrac{u}{2}\big),$$

and

$$a'(u) \cdot \big(\gamma'(u) \times a(u)\big) = -\frac{1}{2}.$$

Therefore

$$\sigma_{uv} \cdot (\sigma_u \times \sigma_v) = -\frac{1}{2}$$

for all (u, v), and hence

$$f = -\frac{1}{2|N|}.$$

Since $F = 0$ and $G = 1$, we know

$$|N|^2 = EG - F^2 = E,$$

so $|N| = \sqrt{E}$ and

$$f = -\frac{1}{2\sqrt{E}}, \qquad g = 0.$$

The Gauss curvature is

$$K = \frac{eg - f^2}{EG - F^2}.$$

Because $g = 0$, we get

$$K = -\frac{f^2}{EG - F^2} = -\frac{\frac{1}{4E}}{E} = -\frac{1}{4E^2}.$$

Since $E(u, v) > 0$ everywhere, it follows that

$$K(u, v) < 0 \quad \text{for all } (u, v) \in [0, 2\pi) \times (-1, 1).$$

Thus the Gaussian curvature of Σ is everywhere negative.

Examples. This surface is a non-developable ruled surface with strictly negative Gaussian curvature. Other examples of ruled surfaces with $K < 0$ include the helicoid and the hyperbolic paraboloid. In contrast, cylinders and cones are ruled surfaces with $K = 0$.

E1 – Strictly convex surfaces and the convex Gauss–Bonnet theorem

Problem. A compact smooth surface $\Sigma \subset \mathbb{R}^3$ is *strictly convex* if it is the boundary of a closed region $R \subset \mathbb{R}^3$ with the property that for any $x, y \in R$, the straight line segment $[x, y] \subset R$, and

$$[x, y] \cap \Sigma \subset \{x, y\}.$$

Show that the Gauss map of a strictly convex surface Σ is a bijection. If, furthermore, the Gauss curvature κ of Σ is everywhere positive, show that the Gauss map is a diffeomorphism, and deduce that

$$\int_\Sigma \kappa \, dA = 4\pi,$$

where κ is the Gauss curvature and dA is the area form on Σ. *(This is the 'convex' Gauss–Bonnet theorem.)*

Theory. For an oriented smooth surface $\Sigma \subset \mathbb{R}^3$ the Gauss map

$$N : \Sigma \to S^2, \qquad N(p) = \text{unit normal at } p,$$

is smooth. Its differential $dN_p : T_p\Sigma \to T_{N(p)}S^2$ is (up to sign) the shape operator, and

$$\det(dN_p) = \kappa(p)$$

is the Gauss curvature. If N is a diffeomorphism then

$$N^*(dA_{S^2}) = \det(dN_p)\, dA = \kappa \, dA,$$

so by change of variables,

$$\int_\Sigma \kappa \, dA = \int_\Sigma N^*(dA_{S^2}) = \int_{S^2} dA_{S^2} = \text{Area}(S^2) = 4\pi.$$

Solution.

Gauss map is a bijection. Fix a unit vector $n \in S^2$ and consider the function

$$h : \Sigma \to \mathbb{R}, \qquad h(x) = x \cdot n.$$

Since Σ is compact, h attains a maximum at some $p \in \Sigma$. At such a maximum the gradient of h is orthogonal to $T_p\Sigma$, so n is parallel to the normal vector at p. Choosing the orientation so that normals point outward, we get $N(p) = n$. Hence N is surjective.

To show injectivity, suppose $N(p_1) = N(p_2) = n$ for some $p_1 \neq p_2$. Let $c = p_1 \cdot n = p_2 \cdot n$ and consider the plane

$$P = \{x \in \mathbb{R}^3 : x \cdot n = c\}.$$

Since R is convex and n is an outward normal, $R \subset \{x \cdot n \leq c\}$, and P is a supporting plane to R. The segment $[p_1, p_2]$ lies entirely in P and in R. Near any point of $[p_1, p_2]$, R lies on one side of P, so every interior point of $[p_1, p_2]$ is on the boundary of R, i.e. belongs to Σ. Thus

$$[p_1, p_2] \cap \Sigma$$

contains more than just the endpoints, contradicting strict convexity. Therefore $p_1 = p_2$, and N is injective.

Hence for a strictly convex surface Σ the Gauss map $N : \Sigma \to S^2$ is bijective.

Gauss map is a diffeomorphism if $\kappa > 0$. We already know N is smooth and bijective. At each $p \in \Sigma$,

$$\det(dN_p) = \kappa(p).$$

If $\kappa(p) > 0$ for all p, then dN_p is invertible for every p. Thus N is a local diffeomorphism.

A smooth bijective local diffeomorphism between compact manifolds of the same dimension is a global diffeomorphism. Consequently, when $\kappa > 0$ everywhere, the Gauss map

$$N : \Sigma \xrightarrow{\cong} S^2$$

is a diffeomorphism.

Integral of Gauss curvature. Let dA_{S^2} denote the area form on the unit sphere. Because N is a diffeomorphism, we have

$$N^*(dA_{S^2}) = \det(dN_p)\, dA = \kappa\, dA.$$

Integrating and using change of variables,

$$\int_\Sigma \kappa\, dA = \int_\Sigma N^*(dA_{S^2}) = \int_{S^2} dA_{S^2} = \text{Area}(S^2) = 4\pi.$$

This proves the convex Gauss–Bonnet formula for strictly convex surfaces with everywhere positive Gauss curvature.

Examples. The surface in the problem is a non–developable ruled surface with strictly negative Gaussian curvature: its parametrisation

$$\sigma(u,v) = \gamma(u) + v\, a(u), \qquad \gamma(u) = (\cos u, \sin u, 0), \quad a(u) = \Big(\cos\frac{u}{2}\cos u, \cos\frac{u}{2}\sin u, \sin\frac{u}{2}\Big),$$

yields Gauss curvature

$$K(u,v) = -\frac{1}{4E(u,v)^2} < 0.$$

Other ruled surfaces with $K < 0$ include:

These examples illustrate that ruled surfaces can have strictly negative Gaussian curvature (helicoid, hyperbolic paraboloid, the example in the problem) or vanishing curvature (cylinders and cones).

1. **Our ruled surface from the problem**

   Chart

   $$\sigma(u,v) = \gamma(u) + v\, a(u), \qquad \gamma(u) = (\cos u, \sin u, 0), \qquad a(u) = \Big(\cos\frac{u}{2}\cos u,\ \cos\frac{u}{2}\sin u,\ \sin\frac{u}{2}\Big).$$

   Each line $v \mapsto \sigma(u,v)$ is a ruling.

   Curvature We computed

   $$K(u,v) = -\frac{1}{4E(u,v)^2} < 0,$$

   so this is a *non-developable* ruled surface with strictly negative Gaussian curvature.

2. **Helicoid – ruled with $K < 0$**

   Chart
   $$\sigma(u, v) = (v \cos u,\ v \sin u,\ au), \qquad u \in \mathbb{R},\ v \in \mathbb{R},\ a > 0.$$
   For fixed u, $v \mapsto \sigma(u, v)$ is a straight line (a radial line in a horizontal plane); as u changes, the line twists upwards – a "spiral ramp".

   Curvature (standard formula)
   $$K(u, v) = -\frac{a^2}{(a^2 + v^2)^2} < 0.$$
   So the helicoid is a classic example of a ruled surface with negative Gauss curvature.

3. **Hyperbolic paraboloid – ruled with $K < 0$**

   Take the saddle surface
   $$z = x\,y.$$
   Chart
   $$\sigma(u, v) = (u,\ v,\ uv).$$
   For fixed $u = u_0$, this is the straight line $(u_0, 0, 0) + v(0, 1, u_0)$; for fixed $v = v_0$, it is $(0, v_0, 0) + u(1, 0, v_0)$. Thus the hyperbolic paraboloid is doubly ruled.

   Curvature
   $$K(u, v) = -\frac{1}{(1 + u^2 + v^2)^2} < 0.$$

4. **Circular cylinder – ruled with $K = 0$**

   Chart
   $$\sigma(u, v) = (\cos u,\ \sin u,\ v), \qquad u \in [0, 2\pi),\ v \in \mathbb{R}.$$
   For fixed u, $v \mapsto \sigma(u, v)$ is a straight vertical line, so the cylinder is ruled.

   Curvature A direct computation gives
   $$K(u, v) = 0.$$
   Cylinders are ruled and *developable* (locally isometric to the plane).

5. **Right circular cone – ruled with $K = 0$ (off the tip)**

   Chart
   $$\sigma(u, v) = (v \cos u,\ v \sin u,\ v), \qquad u \in [0, 2\pi),\ v > 0.$$
   For fixed u, $v \mapsto \sigma(u, v)$ is a straight ray from the apex, so the cone is ruled.

   Curvature Away from the apex one finds
   $$K(u, v) = 0.$$
   The vertex is a singular point (not part of a smooth surface), but on the smooth part the cone is developable.

Ruled, developable and (strictly) convex surfaces

Ruled surfaces. A smooth surface $\Sigma \subset \mathbb{R}^3$ is called *ruled* if through every point $p \in \Sigma$ there passes at least one straight line (a *ruling*) that lies entirely in Σ.

A typical local parametrisation of a ruled surface is

$$\sigma(u, v) = c(u) + v\, d(u),$$

where $c(u)$ is a space curve (the *directrix*) and $d(u)$ is a (usually unit) direction vector. For each fixed u, the map $v \mapsto \sigma(u, v)$ is a straight line.

Examples:

- Plane: $\sigma(u, v) = (u, v, 0)$.
- Circular cylinder: $\sigma(u, v) = (\cos u, \sin u, v)$.
- Cone (away from the tip): $\sigma(u, v) = (v \cos u, v \sin u, v)$.
- Helicoid: $\sigma(u, v) = (v \cos u, v \sin u, au)$.
- Hyperbolic paraboloid: $\sigma(u, v) = (u, v, uv)$.

Thus "ruled" means "built from straight lines"; it does *not* by itself impose any condition on the curvature.

Developable surfaces. A surface Σ is called *developable* if it can be unfolded onto the plane without stretching or tearing, i.e. if it is locally isometric to the plane. For regular surfaces in $\mathbb{R}^3$ this is equivalent to *vanishing Gaussian curvature*:

$$K \equiv 0 \quad \Longleftrightarrow \quad \Sigma \text{ is (locally) developable.}$$

Among ruled surfaces one has the classical classification: a smooth ruled surface with $K = 0$ is locally a piece of a plane, a cylinder, a cone, or a tangent developable of a space curve.

Examples of *developable ruled* surfaces:

- Plane ($K = 0$).
- Circular cylinder ($K = 0$).
- Right circular cone away from the apex ($K = 0$).

Non-developable ruled surfaces (with $K \neq 0$, in fact $K < 0$) include the helicoid, the hyperbolic paraboloid, and the ruled surface from the previous problem.

Convex and strictly convex. A set $R \subset \mathbb{R}^3$ is *convex* if for any $x, y \in R$ the whole segment $[x, y]$ is contained in R:

$$x, y \in R \quad \Rightarrow \quad [x, y] \subset R.$$

A compact smooth surface Σ is called *convex* if it is the boundary of a convex region R.

Typical convex (but not strictly convex) examples: cubes, cylinders, prisms; their boundaries contain flat faces or flat strips.

In the exercise, a compact smooth surface $\Sigma = \partial R$ is called *strictly convex* if R is convex and, in addition,

$$x, y \in R \quad \Rightarrow \quad [x, y] \subset R \quad \text{and} \quad [x, y] \cap \Sigma \subset \{x, y\}.$$

In words: no nontrivial line segment lies on the boundary; the only points of Σ on any chord $[x, y]$ are the endpoints. Thus the boundary has no flat pieces or straight edges: it "bulges outward" everywhere.

Typical strictly convex examples: spheres, ellipsoids, smooth "rounded" convex bodies.

Clearly,

$$\text{strictly convex} \implies \text{convex},$$

but the converse fails (a cylinder is convex but not strictly convex). For smooth surfaces, strict convexity rules out flat regions on Σ; in particular, one expects the principal curvatures to be positive everywhere and hence $K > 0$. In the exercise, the assumptions

$$\Sigma \text{ strictly convex}, \qquad K > 0$$

are used to show that the Gauss map is bijective and has everywhere-invertible differential, hence is a diffeomorphism.

Problem E2 – Second fundamental form and normal curvature

Problem. Let $\Sigma \subset \mathbb{R}^3$ be a smooth oriented surface, and let $p \in \Sigma$. Let $n(p)$ be the unit normal vector at p. Let $v \in T_p\Sigma$ be a unit vector (with respect to the first fundamental form). Let γ_v be the plane curve which is the intersection of Σ with the affine two–plane

$$P = p + \operatorname{span}\bigl(v, n(p)\bigr).$$

Viewing the second fundamental form as a bilinear form

$$\mathrm{II}_p : T_p\Sigma \times T_p\Sigma \to \mathbb{R},$$

show that $\mathrm{II}_p(v, v)$ is the curvature of the plane curve γ_v at p.

(The curvature of a plane curve $\gamma : (a, b) \to \mathbb{R}^2$ parametrized by arc–length is the function $\kappa : (a, b) \to \mathbb{R}$ such that

$$\gamma''(s) = \kappa(s)\, n_\gamma(s),$$

where $n_\gamma(s)$ is the unit normal to γ for which $(\gamma'(s), n_\gamma(s))$ forms a positively oriented basis for $\mathbb{R}^2$.)

Theory. For an oriented surface Σ with unit normal field n, the shape operator $S_p : T_p\Sigma \to T_p\Sigma$ is defined by
$$S_p(w) = -(dn)_p(w).$$
The second fundamental form is the symmetric bilinear form
$$\mathrm{II}_p(u, w) = \langle S_p(u), w\rangle = -\langle (dn)_p(u), w\rangle.$$
If $\alpha : (-\varepsilon, \varepsilon) \to \Sigma$ is a surface curve with $\alpha(0) = p$, $\alpha'(0) = v$ and $\|\alpha'(s)\| = 1$, then
$$\langle \alpha''(0), n(p)\rangle = \mathrm{II}_p(v, v),$$
so $\mathrm{II}_p(v, v)$ is the normal component (in the direction $n(p)$) of the acceleration of α at p.

Solution. Let $\gamma_v : (-\varepsilon, \varepsilon) \to \Sigma$ be the intersection curve of Σ with the plane $P = p + \mathrm{span}(v, n(p))$, parametrized by arc–length so that
$$\gamma_v(0) = p, \qquad \gamma_v'(0) = v, \qquad \|\gamma_v'(s)\| = 1.$$
Since $\gamma_v(s) \in \Sigma$ for all s, we have
$$\big\langle \gamma_v'(s), n\big(\gamma_v(s)\big)\big\rangle = 0 \quad \text{for all } s.$$
Differentiating at $s = 0$ yields
$$\big\langle \gamma_v''(0), n(p)\big\rangle + \big\langle \gamma_v'(0), (dn)_p\big(\gamma_v'(0)\big)\big\rangle = 0,$$
hence, using $\gamma_v'(0) = v$,
$$\big\langle \gamma_v''(0), n(p)\big\rangle = -\big\langle v, (dn)_p(v)\big\rangle = \mathrm{II}_p(v, v),$$
because $\mathrm{II}_p(v, v) = -\langle (dn)_p(v), v\rangle$.

Now consider the curve γ_v as a plane curve in the oriented plane P. We orient P so that $(v, n(p))$ is a positively oriented orthonormal basis. For a plane curve parametrized by arc–length, the curvature at $s = 0$ is the scalar $\kappa_v(0)$ in
$$\gamma_v''(0) = \kappa_v(0)\, n_\gamma(0),$$
where $n_\gamma(0)$ is the unit normal in P such that $(\gamma_v'(0), n_\gamma(0))$ is a positively oriented basis. Since $\gamma_v'(0) = v$ and we have chosen the orientation of P so that $(v, n(p))$ is positively oriented, it follows that $n_\gamma(0) = n(p)$. Therefore
$$\gamma_v''(0) = \kappa_v(0)\, n(p) \quad \Longrightarrow \quad \kappa_v(0) = \big\langle \gamma_v''(0), n(p)\big\rangle.$$
Combining this with the previous identity gives
$$\kappa_v(0) = \mathrm{II}_p(v, v),$$
which proves that the value of the second fundamental form in the direction v equals the curvature of the normal section curve γ_v at p.

Examples.

- For the sphere of radius R, every normal section γ_v is an arc of a great circle of radius R, so its curvature is $1/R$. The shape operator is $S_p = \frac{1}{R}\,\mathrm{Id}$, hence $\mathrm{II}_p(v,v) = \frac{1}{R}$, in agreement with the statement.

- For a plane, the normal field n is constant, so $(dn)_p = 0$ and $\mathrm{II} \equiv 0$. Any normal section is a straight line, whose curvature is 0.

Curvature of a plane curve parametrized by arc–length

In this section we unpack the definition

$$\gamma''(s) = \kappa(s)\, n_\gamma(s),$$

for a plane curve $\gamma : (a,b) \to \mathbb{R}^2$ parametrized by arc–length s, and we illustrate it with examples.

1. What the definition is saying

We have a plane curve

$$\gamma : (a,b) \to \mathbb{R}^2$$

parametrized by *arc–length* s. That means

$$\|\gamma'(s)\| = 1 \quad \text{for all } s.$$

So $\gamma'(s)$ is always a *unit tangent vector*: it tells you the direction of the curve, and its length is 1.

The definition says:

> The *curvature* $\kappa(s)$ is the function such that
>
> $$\gamma''(s) = \kappa(s)\, n_\gamma(s),$$
>
> where $n_\gamma(s)$ is a unit normal vector chosen so that $(\gamma'(s), n_\gamma(s))$ is a *positively oriented* basis of $\mathbb{R}^2$.

Thus:

- $\gamma''(s)$ is the *acceleration* of the point moving along the curve.

- Since $\|\gamma'(s)\| \equiv 1$, one can show that $\gamma''(s)$ is always *orthogonal* to $\gamma'(s)$ (because

$$\frac{d}{ds}\langle \gamma'(s), \gamma'(s)\rangle = 0 \quad \Rightarrow \quad 2\langle \gamma'(s), \gamma''(s)\rangle = 0).$$

- Therefore $\gamma''(s)$ is purely *normal* to the curve: it points in some normal direction and has length $|\kappa(s)|$.

We choose a unit normal $n_\gamma(s)$ (in 2D there are just two choices: "left" or "right"). The condition

$$(\gamma'(s), n_\gamma(s)) \text{ positively oriented}$$

means: in the standard orientation of the plane, if you rotate the tangent vector $\gamma'(s)$ by $+90°$, you get $n_\gamma(s)$. So $n_\gamma(s)$ is the "turn left" normal.

Then we can write

$$\kappa(s) := \langle \gamma''(s), n_\gamma(s) \rangle,$$

and this gives

$$\gamma''(s) = \kappa(s)\, n_\gamma(s).$$

- The *magnitude* $|\kappa(s)|$ tells you how quickly the curve turns.
- The *sign* of $\kappa(s)$ tells you whether the curve is turning towards the chosen normal ("left") or to the opposite side ("right").

2. Curvature as "rate of turning" of the tangent

Because γ is arc–length parametrized, we can write the unit tangent as

$$\gamma'(s) = \big(\cos\theta(s),\, \sin\theta(s)\big),$$

where $\theta(s)$ is the angle that the tangent makes with the x–axis.

Differentiating,

$$\gamma''(s) = \theta'(s)\,(-\sin\theta(s), \cos\theta(s)).$$

But the vector

$$n_\gamma(s) = \big(-\sin\theta(s), \cos\theta(s)\big)$$

is exactly the *left–pointing unit normal* to γ. Thus

$$\gamma''(s) = \theta'(s)\, n_\gamma(s).$$

Comparing with $\gamma''(s) = \kappa(s) n_\gamma(s)$, we obtain

$$\boxed{\kappa(s) = \theta'(s)}.$$

> Curvature at s is the *rate of change of the tangent angle* with respect to arc–length.

If you walk along the curve at unit speed, curvature tells you how fast you are turning your steering wheel per unit distance.

- Straight line: tangent direction is constant, so $\theta'(s) = 0 \Rightarrow \kappa(s) = 0$.
- Circle of radius R traversed counterclockwise: the angle increases at constant rate, $\theta'(s) = 1/R$, so $\kappa(s) = 1/R$.

The *radius of curvature* is

$$\rho(s) = \frac{1}{|\kappa(s)|},$$

the radius of the "best fitting" circle at that point on the curve.

3. Why the sign convention / orientation matters

In the plane we have a fixed notion of positive orientation (usually counterclockwise). At each point of the curve:

- $\gamma'(s)$ points along the curve.
- We pick $n_\gamma(s)$ so that $(\gamma'(s), n_\gamma(s))$ is a positively oriented pair. That is the "turn left" normal.

Then:

- If the curve is turning *left*, $\gamma''(s)$ points in the same direction as $n_\gamma(s)$, so $\kappa(s) > 0$.
- If the curve is turning *right*, $\gamma''(s)$ points opposite to $n_\gamma(s)$, so $\kappa(s) < 0$.

Thus the sign of curvature captures "left vs. right turn".

4. Examples

(a) Straight line. Take

$$\gamma(s) = (s, 0).$$

Then

$$\gamma'(s) = (1, 0), \qquad \gamma''(s) = (0, 0).$$

Hence

$$\gamma''(s) = 0 = 0 \cdot n_\gamma(s), \qquad \kappa(s) \equiv 0.$$

A straight line has zero curvature.

(b) Circle of radius R. Parametrise a circle of radius R counterclockwise by arc–length:

$$\gamma(s) = \big(R\cos(s/R),\ R\sin(s/R)\big).$$

Then

$$\gamma'(s) = \big(-\sin(s/R),\ \cos(s/R)\big),$$

which is a unit tangent vector, and

$$\gamma''(s) = \Big(-\tfrac{1}{R}\cos(s/R),\ -\tfrac{1}{R}\sin(s/R)\Big).$$

The unit normal that makes $(\gamma'(s), n_\gamma(s))$ positively oriented is

$$n_\gamma(s) = \big(-\cos(s/R),\ -\sin(s/R)\big),$$

which points *inward* (towards the center). Then

$$\gamma''(s) = \frac{1}{R}\, n_\gamma(s),$$

so

$$\kappa(s) = \frac{1}{R} > 0.$$

A circle of radius R has constant curvature of magnitude $1/R$, positive when traversed counterclockwise. If we go around the same circle clockwise, the curvature becomes $\kappa = -1/R$.

(c) Parabola $y = x^2$. Consider the parabola

$$\tilde{\gamma}(t) = (t, t^2).$$

This is not arc–length parametrised, but there is a useful general formula for curvature.

For a general parametrisation $\alpha(t) = (x(t), y(t))$,

$$\kappa(t) = \frac{x'(t)y''(t) - y'(t)x''(t)}{\big((x'(t))^2 + (y'(t))^2\big)^{3/2}}.$$

In our case,

$$x'(t) = 1, \quad x''(t) = 0, \quad y'(t) = 2t, \quad y''(t) = 2,$$

so

$$\kappa(t) = \frac{1 \cdot 2 - 2t \cdot 0}{(1 + 4t^2)^{3/2}} = \frac{2}{(1 + 4t^2)^{3/2}} > 0.$$

Thus the parabola bends "left" (in the standard orientation), and its curvature is largest at $t = 0$ (where it is most sharply curved) and decays as $|t| \to \infty$ (far away it looks almost straight). If we reparametrise by arc–length s, we get the same function κ composed with $t = t(s)$, and it still satisfies $\gamma''(s) = \kappa(s)n_\gamma(s)$.

5. Summary insight

- *Arc–length parametrisation* simplifies everything: the tangent vector has unit length, so the acceleration is purely normal.
- The curvature $\kappa(s)$ is exactly the scalar that measures
  - the magnitude of how fast the direction changes per unit distance,
  - the sign telling you whether you turn left or right.
- The equation

  $$\gamma''(s) = \kappa(s)n_\gamma(s)$$

  is the precise mathematical version of the statement:

  > "the acceleration vector points normal to the curve, with size equal to the curvature."

E3 – Fundamental theorem of surface theory in matrix form

Problem. Let $\sigma : V \to \Sigma \subset \mathbb{R}^3$ be an allowable parametrization for a subset of a smooth surface in $\mathbb{R}^3$, where $V \subset \mathbb{R}^2$ is a connected open subset with coordinates (u, v). Write

$$E = \langle \sigma_u, \sigma_u \rangle, \quad F = \langle \sigma_u, \sigma_v \rangle, \quad G = \langle \sigma_v, \sigma_v \rangle$$

for the first fundamental form, and

$$L = \langle \sigma_{uu}, n \rangle, \quad M = \langle \sigma_{uv}, n \rangle, \quad N = \langle \sigma_{vv}, n \rangle$$

for the second fundamental form, where n is the unit normal.

Define

$$\alpha_{ijk} = \langle \sigma_i, \sigma_{jk} \rangle, \qquad i, j, k \in \{u, v\},$$

where subscripts denote partial derivatives.

(a) Show that the functions α_{ijk} on V are determined by the first fundamental form.

(b) Let P be the 3×3 matrix with columns

$$P = (\sigma_u \;\; \sigma_v \;\; n).$$

Define matrix-valued functions on V by

$$C = \begin{pmatrix} E & F & 0 \\ F & G & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad D_1 = \begin{pmatrix} \alpha_{uuu} & \alpha_{uvu} & -L \\ \alpha_{vuu} & \alpha_{vvu} & -M \\ L & M & 0 \end{pmatrix},$$

$$D_2 = \begin{pmatrix} \alpha_{uuv} & \alpha_{uvv} & -M \\ \alpha_{vuv} & \alpha_{vvv} & -N \\ M & N & 0 \end{pmatrix}.$$

Show that

$$P^t P = C, \qquad P^t P_u = D_1, \qquad P^t P_v = D_2.$$

(c) Let $A = C^{-1}D_1$ and $B = C^{-1}D_2$. Deduce that each column ξ of P^t (equivalently, each row of P) satisfies a system of linear first-order partial differential equations

$$\xi_u = A^t \xi, \qquad \xi_v = B^t \xi.$$

Assuming uniqueness of solutions of such a differential system with a given initial condition, deduce that a connected smooth surface in $\mathbb{R}^3$ is determined up to rigid motion by its first and second fundamental forms.

Solution.

(a) The functions α_{ijk} from the first fundamental form. Differentiate

$$E = \langle \sigma_u, \sigma_u \rangle, \quad F = \langle \sigma_u, \sigma_v \rangle, \quad G = \langle \sigma_v, \sigma_v \rangle$$

with respect to u and v. Using the product rule and symmetry of the inner product we obtain

$$\begin{gathered} E_u = 2\langle \sigma_{uu}, \sigma_u \rangle = 2\alpha_{uuu}, \qquad E_v = 2\langle \sigma_{uv}, \sigma_u \rangle = 2\alpha_{uuv}, \\ F_u = \langle \sigma_{uu}, \sigma_v \rangle + \langle \sigma_u, \sigma_{uv} \rangle = \alpha_{vuu} + \alpha_{uvu}, \\ F_v = \langle \sigma_{uv}, \sigma_v \rangle + \langle \sigma_u, \sigma_{vv} \rangle = \alpha_{vuv} + \alpha_{uvv}, \\ G_u = 2\langle \sigma_{uv}, \sigma_v \rangle = 2\alpha_{vuv}, \qquad G_v = 2\langle \sigma_{vv}, \sigma_v \rangle = 2\alpha_{vvv}. \end{gathered}$$

Hence

$$\alpha_{uuu} = \frac{1}{2}E_u, \qquad \alpha_{uuv} = \frac{1}{2}E_v, \qquad \alpha_{vuv} = \frac{1}{2}G_u, \qquad \alpha_{vvv} = \frac{1}{2}G_v.$$

By symmetry of mixed partial derivatives and the inner product, $\alpha_{vuu} = \alpha_{uvu}$ and $\alpha_{uvv} = \alpha_{vuv}$, so from the equations for F_u and F_v we also get

$$\alpha_{vuu} = \alpha_{uvu} = \frac{1}{2}F_u, \qquad \alpha_{uvv} = \alpha_{vuv} = \frac{1}{2}F_v.$$

Thus all α_{ijk} are expressed in terms of E, F, G and their first derivatives; they are determined by the first fundamental form.

(b) The matrices C, D_1, D_2. The (i,j)-entry of P^tP is the inner product of the i-th and j-th columns of P, i.e.

$$P^tP = \begin{pmatrix} \langle \sigma_u, \sigma_u\rangle & \langle \sigma_u, \sigma_v\rangle & \langle \sigma_u, n\rangle \\ \langle \sigma_v, \sigma_u\rangle & \langle \sigma_v, \sigma_v\rangle & \langle \sigma_v, n\rangle \\ \langle n, \sigma_u\rangle & \langle n, \sigma_v\rangle & \langle n, n\rangle \end{pmatrix}.$$

Since n is a unit normal, we have $\langle \sigma_u, n\rangle = \langle \sigma_v, n\rangle = 0$ and $\langle n, n\rangle = 1$, so

$$P^tP = \begin{pmatrix} E & F & 0 \\ F & G & 0 \\ 0 & 0 & 1 \end{pmatrix} = C.$$

Next, P_u has columns σ_{uu}, σ_{uv} and n_u, so

$$P^tP_u = \begin{pmatrix} \langle \sigma_u, \sigma_{uu}\rangle & \langle \sigma_u, \sigma_{uv}\rangle & \langle \sigma_u, n_u\rangle \\ \langle \sigma_v, \sigma_{uu}\rangle & \langle \sigma_v, \sigma_{uv}\rangle & \langle \sigma_v, n_u\rangle \\ \langle n, \sigma_{uu}\rangle & \langle n, \sigma_{uv}\rangle & \langle n, n_u\rangle \end{pmatrix}.$$

The first 2×2 block is

$$\alpha_{uuu},\ \alpha_{uvu},\ \alpha_{vuu},\ \alpha_{vvu}$$

by definition. To compute the last column, note that $\langle \sigma_u, n\rangle = 0$ implies

$$0 = \frac{\partial}{\partial u}\langle \sigma_u, n\rangle = \langle \sigma_{uu}, n\rangle + \langle \sigma_u, n_u\rangle = L + \langle \sigma_u, n_u\rangle,$$

so $\langle \sigma_u, n_u\rangle = -L$. Similarly,

$$0 = \frac{\partial}{\partial u}\langle \sigma_v, n\rangle = \langle \sigma_{vu}, n\rangle + \langle \sigma_v, n_u\rangle = M + \langle \sigma_v, n_u\rangle$$

gives $\langle \sigma_v, n_u\rangle = -M$, while $\langle n, n\rangle = 1$ yields $\langle n, n_u\rangle = 0$.

For the last row we have

$$\langle n, \sigma_{uu}\rangle = L, \qquad \langle n, \sigma_{uv}\rangle = M, \qquad \langle n, n_u\rangle = 0.$$

Hence

$$P^tP_u = D_1.$$

The proof that $P^tP_v = D_2$ is analogous, using $P_v = (\sigma_{uv}\ \ \sigma_{vv}\ \ n_v)$ and

$$\langle \sigma_u, n_v\rangle = -M, \quad \langle \sigma_v, n_v\rangle = -N, \quad \langle n, n_v\rangle = 0.$$

(c) The PDE system and rigidity. From (b),

$$P^tP = C, \qquad P^tP_u = D_1, \qquad P^tP_v = D_2.$$

Define

$$A = C^{-1}D_1, \qquad B = C^{-1}D_2.$$

Then

$$A = C^{-1}D_1 = C^{-1}P^tP_u.$$

Using $C = P^t P$ we observe

$$C^{-1}P^t = (P^t P)^{-1}P^t = P^{-1},$$

so

$$A = P^{-1}P_u \quad \Rightarrow \quad P_u = PA.$$

Similarly $B = P^{-1}P_v$ and hence $P_v = PB$.

Taking transposes of these equations gives

$$P_u^t = A^t P^t, \qquad P_v^t = B^t P^t.$$

Let ξ be any column of P^t (equivalently, the transpose of a row of P). Then the above equations say precisely

$$\xi_u = A^t\xi, \qquad \xi_v = B^t\xi,$$

so each such ξ satisfies a linear first-order system with coefficients determined by the first and second fundamental forms.

Now suppose Σ_1 and Σ_2 are two connected smooth surfaces with the same first and second fundamental forms on a common parameter domain V. Then the associated matrices C, D_1, D_2 and hence A, B coincide for both. Let $P^{(1)}$ and $P^{(2)}$ be the corresponding matrices whose columns are σ_u, σ_v, n for each surface. The columns of $(P^{(1)})^t$ and $(P^{(2)})^t$ satisfy the same PDE system

$$\xi_u = A^t\xi, \qquad \xi_v = B^t\xi.$$

Choose $(u_0, v_0) \in V$ and, by applying a rigid motion to Σ_2 if necessary, arrange that

$$\sigma^{(1)}(u_0, v_0) = \sigma^{(2)}(u_0, v_0), \qquad P^{(1)}(u_0, v_0) = P^{(2)}(u_0, v_0).$$

Thus the initial values of the corresponding columns of $(P^{(1)})^t$ and $(P^{(2)})^t$ agree at (u_0, v_0). By the assumed uniqueness of solutions for the linear system with these initial conditions, it follows that

$$P^{(1)}(u, v) = P^{(2)}(u, v)$$

for all (u, v) in the connected set V. In particular, $\sigma_u^{(1)} = \sigma_u^{(2)}$ and $\sigma_v^{(1)} = \sigma_v^{(2)}$, so the two parametrizations differ only by a fixed rigid motion (constant orthogonal transformation and translation) in $\mathbb{R}^3$.

Therefore a connected smooth surface in $\mathbb{R}^3$ is determined, up to rigid motion, by its first and second fundamental forms.

Problem 1: Geodesics on the unit cylinder

Problem statement. Let

$$\Sigma = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 = 1\}$$

be the unit cylinder. Show that a geodesic on Σ through the point $(1, 0, 0)$ can be parametrized so that it is contained in a spiral of the form

$$\gamma(t) = (\cos\alpha t,\ \sin\alpha t,\ \beta t), \qquad \alpha^2 + \beta^2 = 1.$$

Theory. The map

$$F(\theta, z) = (\cos\theta, \sin\theta, z), \qquad (\theta, z) \in \mathbb{R}^2$$

parametrizes the cylinder. Its partial derivatives are

$$F_\theta = (-\sin\theta, \cos\theta, 0), \qquad F_z = (0, 0, 1),$$

so the induced metric on Σ is

$$g = d\theta^2 + dz^2.$$

Hence F is a local isometry from the flat plane $(\mathbb{R}^2, d\theta^2 + dz^2)$ onto Σ. Isometries send geodesics to geodesics (up to reparametrization), and geodesics in the Euclidean plane are straight lines.

Solution. The point $(1, 0, 0)$ corresponds to $(\theta, z) = (0, 0)$. In the (θ, z)-plane, all unit–speed geodesics through $(0, 0)$ are straight lines of the form

$$\widetilde{\gamma}(t) = (\alpha t,\ \beta t), \qquad \alpha^2 + \beta^2 = 1,$$

since $|\dot{\widetilde{\gamma}}|^2 = \alpha^2 + \beta^2$. Applying F we obtain a curve on the cylinder:

$$\gamma(t) = F(\widetilde{\gamma}(t)) = F(\alpha t, \beta t) = (\cos(\alpha t),\ \sin(\alpha t),\ \beta t).$$

We have $\gamma(0) = (\cos 0, \sin 0, 0) = (1, 0, 0)$, so the curve passes through the required point. Because F is an isometry and $\widetilde{\gamma}$ is a unit–speed straight line, γ is a unit–speed geodesic on Σ. Its image is a spiral around the cylinder, with constant angular velocity $|\alpha|$ and constant vertical velocity $|\beta|$. Thus every geodesic through $(1, 0, 0)$ can be reparametrized into the form

$$\gamma(t) = (\cos\alpha t,\ \sin\alpha t,\ \beta t), \qquad \alpha^2 + \beta^2 = 1.$$

Examples / comments.

- For $\alpha = 0$, $\beta = \pm 1$ we get the vertical generator $\gamma(t) = (1, 0, \pm t)$, which is a geodesic.
- Spirals with rational slope $\beta/\alpha \in \mathbb{Q}$ close up to form closed geodesics on the cylinder; for irrational slope they wind around densely.

Problem 2: Straight lines as geodesics and the hyperboloid

Problem statement. Let $\Sigma \subset \mathbb{R}^3$ be a smooth embedded surface. Suppose that a straight line $\ell = \mathbb{R} \subset \mathbb{R}^3$ lies entirely in Σ.

1. Prove that ℓ is a geodesic on Σ.
2. Deduce that through every point p of the hyperboloid

$$S = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 = z^2 + 1\}$$

there are (at least) three geodesics $\gamma_p : \mathbb{R} \to S$ defined on the entire real line $\mathbb{R}$.

Theory. For a smooth curve $\gamma : I \to \Sigma \subset \mathbb{R}^3$, its acceleration vector in $\mathbb{R}^3$ decomposes into tangential and normal components:

$$\gamma''(t) = (\nabla^{\Sigma}_{\dot\gamma}\dot\gamma)(t) + (\text{normal component}).$$

The intrinsic covariant derivative $\nabla^{\Sigma}_{\dot\gamma}\dot\gamma$ is tangent to Σ, and the geodesic equation is

$$\nabla^{\Sigma}_{\dot\gamma}\dot\gamma = 0,$$

i.e. the tangential part of γ'' vanishes.

A straight line in $\mathbb{R}^3$ has zero acceleration, $\gamma'' \equiv 0$. Moreover, the hyperboloid S is a surface of revolution about the z-axis, so its meridians (curves with constant polar angle θ) are geodesics. From a previous result we know that S is doubly ruled: through each point of S there pass two distinct straight lines lying entirely in S.

Solution (1). Let $\ell \subset \Sigma$ be a straight line and parametrize it as

$$\gamma(t) = p_0 + tv, \qquad t \in \mathbb{R},$$

for some fixed $p_0 \in \Sigma$ and constant $v \in \mathbb{R}^3 \setminus \{0\}$. Then

$$\gamma'(t) = v, \qquad \gamma''(t) = 0.$$

Decomposing into tangential and normal components along Σ,

$$0 = \gamma''(t) = (\nabla^{\Sigma}_{\dot\gamma}\dot\gamma)(t) + (\text{normal component}).$$

Therefore $\nabla^{\Sigma}_{\dot\gamma}\dot\gamma \equiv 0$, so γ satisfies the geodesic equation on Σ. Hence ℓ is a geodesic, and the parametrization is defined for all $t \in \mathbb{R}$.

Solution (2). Let $p \in S$.

- By the ruled-surface property of the hyperboloid, there exist two distinct straight lines $\ell_1, \ell_2 \subset S$ passing through p. By part (1), each ℓ_i is a geodesic on S, and with its natural affine parameter it is defined for all $t \in \mathbb{R}$.
- Since S is a surface of revolution about the z-axis, the meridian through p (intersection with the plane through the axis containing p) is also a geodesic. In cylindrical coordinates p has the form (r, θ_0, z_0) with $r = \sqrt{z_0^2 + 1}$, and the meridian can be parametrized as

  $$\gamma_p(t) = \big(\sqrt{1+t^2}\cos\theta_0,\ \sqrt{1+t^2}\sin\theta_0,\ t\big), \qquad t \in \mathbb{R}.$$

  This parametrization is defined on all of $\mathbb{R}$.

Thus through every point $p \in S$ there are at least three distinct complete geodesics: the two straight ruling lines and the meridian.

Examples / comments. At the point $p = (1, 0, 0)$ the meridian geodesic is

$$\gamma(t) = (\sqrt{1+t^2}, 0, t),$$

and there are two ruling lines through p, belonging to the two distinct families that rule the hyperboloid.

Definition 19 (Geodesic). *Let (Σ, g) be a Riemannian surface with Levi–Civita connection ∇. A smooth curve $\gamma : I \to \Sigma$ is called a* geodesic *if its velocity vector is parallel along the curve, i.e.*

$$\nabla_{\dot\gamma(t)}\dot\gamma(t) = 0 \quad \text{for all } t \in I.$$

Equivalently, the geodesic curvature of γ is zero, or in local coordinates (u^1, u^2) with Christoffel symbols Γ^k_{ij} the components $u^k(t)$ of γ satisfy

$$\ddot u^k + \Gamma^k_{ij}\,\dot u^i \dot u^j = 0.$$

Theorem 0.5 (Geodesic through a point in a given direction). *Let (Σ, g) be a Riemannian surface. For every point $p \in \Sigma$ and every tangent vector $v \in T_p\Sigma$ there exist an open interval $I \subset \mathbb{R}$ with $0 \in I$ and a unique geodesic $\gamma : I \to \Sigma$ such that*

$$\gamma(0) = p, \qquad \dot\gamma(0) = v.$$

If Σ is geodesically complete, this geodesic extends to all $t \in \mathbb{R}$.

Definition 20 (Surface of revolution, meridians and parallels). *Let $(r(u), z(u))$ be a plane curve in the rz-plane with $r(u) > 0$. Rotating it about the z-axis gives a surface of revolution with parametrization*

$$X(u, v) = \big(r(u)\cos v,\ r(u)\sin v,\ z(u)\big), \quad u \in I,\ v \in \mathbb{R}.$$

Curves with $v = \text{const}$ are called meridians*, and curves with $u = \text{const}$ are called* parallels.

Proposition 0.6 (Meridians are geodesics). *On a surface of revolution parametrized as above, the induced metric has the form*

$$g = E(u)\,du^2 + G(u)\,dv^2, \qquad E(u) = r'(u)^2 + z'(u)^2, \quad G(u) = r(u)^2.$$

For each fixed v_0, the meridian

$$\gamma(u) = X(u, v_0)$$

satisfies the geodesic equations and hence is a geodesic on the surface.

Example (Meridian geodesics on the hyperboloid). The hyperboloid

$$S = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 = z^2 + 1\}$$

is a surface of revolution. In cylindrical coordinates we can write $r(z) = \sqrt{1 + z^2}$, and for any fixed angle θ_0 the meridian

$$\gamma(t) = \big(\sqrt{1+t^2}\cos\theta_0,\ \sqrt{1+t^2}\sin\theta_0,\ t\big), \quad t \in \mathbb{R},$$

is a geodesic on S.

Definition 21 (Geodesic as locally shortest path). *Let (Σ, g) be a Riemannian surface. A smooth curve $\gamma : [a, b] \to \Sigma$ is called a* geodesic *if it is locally length-minimizing: for every $t_0 \in (a, b)$ there exists $\varepsilon > 0$ such that for any smooth curve η with*

$$\eta(t_0 - \varepsilon) = \gamma(t_0 - \varepsilon), \qquad \eta(t_0 + \varepsilon) = \gamma(t_0 + \varepsilon),$$

we have

$$L(\eta) \ \geq \ L\big(\gamma|_{[t_0-\varepsilon, t_0+\varepsilon]}\big).$$

Equivalently, γ has zero geodesic curvature or satisfies the geodesic equation $\nabla_{\dot\gamma}\dot\gamma = 0$.

Definition 22 (Levi–Civita connection)**.** *Let* (Σ, g) *be a Riemannian surface. A* connection *(or covariant derivative) is an operator* ∇ *assigning to vector fields* X, Y *a new vector field* $\nabla_X Y$ *such that*

1. $\nabla_{fX+gY} Z = f\nabla_X Z + g\nabla_Y Z,$
2. $\nabla_X(fY) = X(f)Y + f\nabla_X Y,$

for smooth functions f, g*. The connection is called* torsion-free *if* $\nabla_X Y - \nabla_Y X = [X, Y]$ *and* metric-compatible *if*

$$X\big(g(Y, Z)\big) = g(\nabla_X Y, Z) + g(Y, \nabla_X Z)$$

for all vector fields X, Y, Z*. The unique torsion-free, metric-compatible connection on* (Σ, g) *is called the* Levi–Civita connection*.*

Definition 23 (Christoffel symbols)**.** *Let* (u^1, u^2) *be local coordinates on* Σ *and let* $\partial_i = \frac{\partial}{\partial u^i}$*. The* Christoffel symbols *of the Levi–Civita connection are the functions* Γ^k_{ij} *defined by*

$$\nabla_{\partial_i} \partial_j = \Gamma^k_{ij}\, \partial_k.$$

If g_{ij} *are the components of the metric and* (g^{ij}) *is its inverse matrix, then*

$$\Gamma^k_{ij} = \frac{1}{2}\, g^{k\ell}\big(\partial_i g_{j\ell} + \partial_j g_{i\ell} - \partial_\ell g_{ij}\big).$$

Proposition 0.7 (Geodesic equation in coordinates)**.** *Let* $\gamma(t) = (u^1(t), u^2(t))$ *be a curve in local coordinates. Then* γ *is a geodesic if and only if it satisfies*

$$\ddot{u}^k(t) + \Gamma^k_{ij}\big(u(t)\big)\, \dot{u}^i(t)\, \dot{u}^j(t) = 0, \qquad k = 1, 2.$$